Third Asia-Pacific consensus recommendations on colorectal cancer screening and postpolypectomy surveillance

Joseph J Y Sung ,[1] Han-Mo Chiu ,[2,3] David Lieberman,[4] Ernst J Kuipers ,[5] Matthew D Rutter ,[6] Finlay Macrae,[7] Khay-Guan Yeoh ,[8] Tiing Leong Ang,[9] Vui Heng Chong,[10] Sneha John ,[11] Jingnan Li,[12] Kaichun Wu,[13] Simon S M Ng,[14] Govind K Makharia ,[15] Murdani Abdullah,[16,17] Nozomu Kobayashi ,[18,19] Masau Sekiguchi ,[18,19] Jeong-Sik Byeon ,[20] Hyun-Soo Kim ,[21] Susan Parry,[22,23] Patricia Anne I Cabral-Prodigalidad,[24] Deng-Chyang Wu,[25] Suparkij Khomvilai,[26] Rashid N Lui ,[27,28] Sunny Wong ,[1] Yu-Min Lin,[29] E Dekker [30]

For numbered affiliations see end of article.

Correspondence to
Professor Joseph J Y Sung, Lee Kong Chian School of Medicine, Singapore, Singapore; josephsung@ntu.edu.sg

JJYS and H-MC are joint first authors.

Received 15 March 2022
Accepted 7 August 2022

Check for updates

© Author(s) (or their employer(s)) 2022. No commercial re-use. See rights and permissions. Published by BMJ.

To cite: Sung JJY, Chiu H-M, Lieberman D, *et al*. *Gut* Epub ahead of print: [*please include* Day Month Year]. doi:10.1136/gutjnl-2022-327377

ABSTRACT

The Asia-Pacific region has the largest number of cases of colorectal cancer (CRC) and one of the highest levels of mortality due to this condition in the world. Since the publishing of two consensus recommendations in 2008 and 2015, significant advancements have been made in our knowledge of epidemiology, pathology and the natural history of the adenoma-carcinoma progression. Based on the most updated epidemiological and clinical studies in this region, considering literature from international studies, and adopting the modified Delphi process, the Asia-Pacific Working Group on Colorectal Cancer Screening has updated and revised their recommendations on (1) screening methods and preferred strategies; (2) age for starting and terminating screening for CRC; (3) screening for individuals with a family history of CRC or advanced adenoma; (4) surveillance for those with adenomas; (5) screening and surveillance for sessile serrated lesions and (6) quality assurance of screening programmes. Thirteen countries/regions in the Asia-Pacific region were represented in this exercise. International advisors from North America and Europe were invited to participate.

INTRODUCTION

The Asia-Pacific region contributes the largest burden of cases of colorectal cancer (CRC) (51.8%) and CRC mortality (52.4%) in the world.[1] Countries in this region, including Australia, New Zealand, China, Japan and Korea, have some of the highest CRC incidences in the world. The rising number may be related to the rapid economic growth in this region as the incidence of CRC rises in parallel with economic transition from a low to high Human Development Index (HDI),[2] a composite index of life expectancy, education and per capita income.

Since we published the two Asia-Pacific Consensus Recommendations on Colorectal Cancer Screening in 2008[3] and 2015,[4] significant advancements have been made in the knowledge of the epidemiology, pathology and natural history of the adenoma-carcinoma progression. Emerging evidence shows a trend for CRC to be diagnosed in younger patients, below the age of 50 years. Screening programmes starting at the age of 50 years may therefore not be adequate. New stool-based or blood-based molecular markers have been developed providing more non-invasive choices beyond testing for faecal occult blood. Risk stratification for CRC and whether a one-stop screening or sequential offering of tests are optimal strategies remain a subject of debate. More studies on the risk of CRC with family history of the malignancy or advanced adenoma have been published in recent years. With these advancements, recommendations for screening strategies should be revised. Surveillance methods and intervals after resection of colonic adenomas also need to be re-examined.

In the Asia-Pacific region, little work has been done on surveillance strategies until recently. Important data arising from clinical studies from Taiwan and Japan shed new light on this important issue. Further, sessile serrated lesions (SSLs) were rarely reported in Asian studies. The reported low prevalence could be due to low awareness and underdiagnosis of these lesions. Therefore, guidelines on the surveillance strategy and management of SSLs are very much needed for this region. Although CRC screening programmes, organised or opportunistic, are mushrooming in Asia, there is a lack of guidance on the quality assurance of these programmes. Despite an increase in screening programmes in Asia, there is little information about adherence to, and quality of, these programmes.

In view of these developments, the Asia-Pacific Working Group on Colorectal Cancer Screening decided to convene a further consensus in 2019 with the aim offering an update on CRC screening recommendations, especially to cover areas not dealt with in the previous two consensus recommendations.[3 4] Originally, it was planned to be a 2-day in-person meeting but because of the coronavirus disease (COVID-19) pandemic, the consensus

meeting was postponed and eventually conducted as a virtual meeting.

METHOD

The consensus was conducted as a virtual meeting. The process was carefully designed to adhere strictly to the modified Delphi process, similar to the last two consensus conferences in 2008 and 2015.

Membership of the consensus panel

Members of the consensus group were selected based on the following criteria: (1) demonstrated knowledge/expertise in CRC by publication/research or participation in development of national or regional guidelines; (2) geographical representation of various Asia-Pacific countries/regions; (3) participation in the Asia-Pacific Working Group for CRC screening research projects and/or previous Asia-Pacific Consensus Recommendations process in 2014. International experts who had played key roles in drafting other regional/national/international guidelines were invited to contribute to the drafting, discussion and voting for these recommendations. Membership was divided into four categories: (1) coordinators (JJYS, H-MC, ED) were responsible for drafting the statements and coordinating the process of discussion and voting on the statements; (2) advisors (DL, EJK, MDR, FM) helped to draft the first statements, moderate discussion during the meeting and revise the statements accordingly; (3) panels (K-GY, TLA, VHC, SJ, JL, KW, SSMN, GKM, MA, NK, MS, J-SB, H-SK, SP, PAIC-P, D-CW, SK) were country/region representatives who participated in the discussions and voting of the statements during the virtual conference; (4) facilitators (SW, RNL, Y-ML) were responsible for performing the literature search, collating the discussion points and revising the statements.

Provisional statements

The consensus is grouped under six topics (1) screening methods and preferred strategies; (2) age for starting and terminating screening for CRC; (3) screening for individuals with a family history of CRC or advanced adenomas; (4) surveillance for those with adenomas; (5) screening and surveillance for SSLs and (6) quality assurance of screening programmes. The consensus recommendation statements were drafted by the coordinators and sent to the advisors for comments. The first version of the statements was then sent to the panels for voting, which they carried out based on evidence in the literature provided by the facilitators.

Literature search

A comprehensive literature search was conducted by the facilitators. Relevant articles published in the English language up to September 2021, using EMB Reviews, Embase, Ovid Medline, PubMed Medline, Cochrane Library and Google Scholar, were included. Searches were performed using the following key words: colorectal cancer, advanced adenoma, cancer screening, family history, adenoma detection rate, advanced neoplasia, Asia, colonoscopy, colorectal polyps, cost-effectiveness, CT colonography, diabetes, early detection, faecal immunochemical test, guidelines; interval cancer, metabolic syndrome, obesity, opportunistic screening, optical digital method, quality indicator, personalised screening, polypectomy, population-based screening, post-colonoscopy colorectal cancer, primary prevention, risk stratification, sessile serrated adenoma/polyp, sessile serrated lesions, screening, sigmoidoscopy, stool DNA, surveillance. National and international guidelines on CRC screening were also solicited. Our initial search identified 319 full papers.

Voting process

The working group started the first round of voting after receiving the first draft of the statements. Based on the literature and the panel members' knowledge and opinion, anonymous votes were submitted online. A Likert scale anchored by 1–5 (A=accept completely, B=accept with some reservation, C=accept with major reservation, D=reject with reservation, E=reject completely) was adopted. Consensus was achieved when >80% of the votes indicated 'accept completely' or 'accept with some reservation.' A statement was refuted when >80% of the votes indicated 'reject completely' or 'reject with reservation'. Statements for which a consensus could not be reached were discussed and modified at the 2-day conference. During the consensus conference, seven panel members (JJYS, H-MC, ED, DL, EJK, MDR, FM) were invited to review and present articles searched from the literature in each of the six topics (see above) before introducing each of the consensus statement for voting. Then a second round of voting was conducted. If consensus was still not reached, the statement was modified for the last time. A third and the last round of voting was then conducted leading to final acceptance or refutation. Each statement was then graded to indicate the level of evidence available and to indicate the strength of recommendation (table 1). Each voting member signed a declaration of conflict of interests on screening strategy and financial relationship to industry.

RESULTS

A 2-day online consensus conference was held on 27–28 November 2021. Representatives from 13 Asia-Pacific countries/regions participated in the meeting, including Australia, Brunei, China, Hong Kong, India, Indonesia, Korea, Japan, New Zealand, Philippines, Singapore, Taiwan and Thailand. Twenty-four members took part in the voting.

Screening methods and strategies of choice

Statement 1. Risk stratification and sequential offering of tests are reasonable approaches to CRC screening in Asia.

Level of agreement: A=54.1%, B=41.7%, C=4.2%, D=0%, E=0%

Quality of evidence: I

Classification of recommendation: A

The incidence of CRC increases significantly after the age of 50 in most of the developed countries. Offering direct colonoscopy screening for the entire eligible population above 50 years of age is, however, not feasible in the majority of the countries in Asia, where the single-payer healthcare system is the mainstay, with constraints in funding, clinical capacity and manpower, and where public awareness of CRC and its screening is still low.[1] Moreover, considering the prevalence of 2 to 3 per 1000 for CRC and 3–10% for advanced adenoma in the general screening population, a risk-based stratification and sequential offering of a screening test is a more pragmatic and efficient approach.[5] A faecal immunochemical test (FIT) is currently the most popular screening test used in organised screening programmes, including the Asia-Pacific region.[5] The positivity rate of a FIT, which is largely affected by the population risk of colorectal neoplasm and the FIT cutoff value used to determine positivity (based, for example, on the colonoscopy capacity in individual programmes), is generally within the range of 4% to

Table 1 Voting, quality of evidence and classification of recommendations

Category and grade	Description
Voting on recommendations	
A	Accept completely
B	Accept with some reservation
C	Accept with major reservation
D	Reject with some reservation
E	Reject completely
Quality of evidence	
I	Evidence obtained from at least one RCT
II-1	Evidence obtained from well-designed control trials without randomisation
II-2	Evidence obtained from well-designed cohort or case–control study
II-3	Evidence obtained from comparison between time or places with or without intervention
III	Opinion of respected authorities, based on clinical experience and expert committees
Classification of recommendations	
A	There is good evidence to support the statement
B	There is fair evidence to support the statement
C	There is poor evidence to support the statement but recommendation made on other grounds
D	There is fair evidence to refute the statement
E	There is good evidence to refute the statement

RCT, randomised controlled trial.

10%. In a recent Polish study in which 12 485 individuals were invited to either direct colonoscopy screening or a sequential FIT/colonoscopy screening or a choice between FIT and colonoscopy screening, the participation rate was found to be significantly higher in the sequential or choice strategy.[6] A randomised controlled trial from China has provided strong evidence that a risk-adapted screening strategy with colonoscopy and FIT in colorectal cancer screening can achieve much higher participation.[7] Therefore, using risk stratification followed by colonoscopy if the test is positive, will significantly improve screening uptake and reduce the colonoscopy burden of a screening programme.

Statement 2. Multiple risk-stratification systems have been developed and no single stratification method has proved to be better than the others

Level of agreement: A=83.3%, B=16.7%, C=0%, D=0%, E=0%

Quality of evidence: II-2

Classification of recommendation: A

In the Asia-Pacific region, FIT is the most popular test for identifying individuals in the general population at high risk for advanced neoplasia (AN). In countries with rising CRC incidence but with no organised screening programmes yet in place, risk-score systems using simple demographics, family history or other CRC risk factors for triaging whether these individuals warrant a colonoscopy would be good alternatives to FIT and could help to make the most efficient use of the constrained endoscopy resources. The Asia-Pacific Risk Score was developed and validated in 15 Asia-Pacific countries/regions.[8–11] Using this risk-score system, subjects in the middle- and high-risk tiers had 2.6-fold and 4.3-fold increased prevalence of AN/CRC, respectively. When used in combination with FIT, the risk score could detect around 70% of subjects with AN and 95% of subjects with cancers, both significantly higher than FIT screening alone, and hence reducing colonoscopy demand by half compared with direct colonoscopy screening.[12] Several other risk-score systems were also developed, both in Asia and in other continents, using similar risk factors such as age, gender, family history, smoking, drinking and body mass index in different combinations and weighting. No single risk-score system was proved to be better than the others in head-to-head comparison.[13] As the American Gastroenterological Association (AGA) white paper stated, "a one-size-fits-all approach to CRC screening has not and is unlikely to result in increased screening uptake or desired outcomes owing to barriers stemming from behavioural, culture and socioeconomic technologies".[14] Six centres from China including 19 546 subjects also showed that a risk-adapted screening strategy (ie, high-risk subjects for colonoscopy and low-risk subjects for a FIT) had a much better participation rate.[7] In the Asia-Pacific region, there is a broad variety of disease prevalence, healthcare funding models, capacity and manpower issues, which may affect the choice of risk stratification approach and selection of CRC screening strategies. The Asia-Pacific Working Group recommends that any of these risk stratification systems should be adopted after validation in local regions.

Statement 3. Quantitative faecal immunochemical test (every year or every 2 years) or colonoscopy (every 10 years) are the preferred screening tests in Asia.

Level of agreement: A=45.9%, B=45.8%, C=8.3%, D=0%, E=0%

Quality of evidence: II-2

Classification of recommendation: A

Accumulating evidence demonstrates the effectiveness of a FIT and screening colonoscopy to reduce CRC incidence and death.[15–19] The time intervals for FIT and colonoscopy are based on large-scale, long-duration, follow-up cohort studies. The use of an annual faecal occult blood test significantly reduces the incidence of colorectal cancer.[20] The effect of faecal occult blood testing on CRC is long lasting. The reduced mortality related to CRC is found to persist after 30 years of follow up.[21] A simulation model of CRC screening by the US Preventive Services Task Force (USPSTF) also confirmed that 1–2 yearly screening is better than 3-yearly screening.[22] A retrospective cohort study from northern California that included over 1 million individuals confirmed that a negative colonoscopy results in reduced risk of colorectal cancer and related death for more than 12 years after examination compared with unscreened patients.[23] A Polish observational study also confirmed that after a single negative screening colonoscopy, profound and stable reduction in CRC incidence and mortality was documented for up to 17 years.[24]

In the Asia-Pacific region, organised nationwide screening programmes are in place in Australia, Brunei, Hong Kong, Japan, South Korea, New Zealand, Singapore and Taiwan, and all these programmes use FIT as the primary screening test. In the opportunistic screening settings, colonoscopy is more often used as primary screening tool.[5] A recent cohort study with 10-year follow-up from Taiwan revealed that biennial FIT screening could significantly reduce advanced stage CRC incidence by 29% and CRC mortality by 35%.[16] Flexible sigmoidoscopy is currently not used in organised screening programmes in this region, although its effectiveness in reducing CRC incidence and mortality in the distal colon is demonstrated by randomised trials. This is probably related to increased accessibility to colonoscopy and the concern about increasing prevalence of proximal colorectal neoplasms.[25–28] Guaiac-based faecal occult blood tests are no longer used and have been largely replaced by FIT owing to its convenience of use, high screening uptake, and better sensitivity and specificity for CRC. Therefore, the working group recommends quantitative FIT (every year or every 2 years) or colonoscopy (every 10 years) as the preferred screening tests in Asia.

Statement 4. There are insufficient data for stool- or blood-based molecular tests and capsule colonoscopy and CT colonoscopy in Asia to be recommended as a primary screening tool.

Level of agreement: A=87.5%, B=12.5%, C=0%, D=0%, E=0%

Quality of evidence: III

Classification of recommendation: A

A multitarget stool-based DNA test was developed in the United States and approved by the Food and Drug Administration. Its sensitivity in detecting CRC and adenomas with high-grade dysplasia was 92.3% and 69.2%, respectively, both significantly higher than with a FIT.[29] A recent comprehensive economic appraisal comparing different CRC screening strategies also showed that its cost-effectiveness may be comparable to that of annual FIT screening.[22] There are, however, still significant concerns about using it for large-scale population screening owing to the unclear screening uptake by the public, inconvenience of stool collection and shipping, high cost and the lack of evidence on its programme sensitivity demonstrating its superiority over FIT.[22] To date, no validation studies on multitarget stool-based DNA testing have been conducted in Asia. Blood-based tests using cell-free DNA, proteins, messenger RNA, micro-RNA, or even circulating tumour cells as the target biomarkers were also developed, and some of the studies have shown promising results but validation studies in large screening populations are still lacking which hinders their roles as a front-line screening test at this moment.[30]

In a recent meta-analysis that included studies conducted in screening populations, capsule colonoscopy detected 93% of the CRCs (using colonoscopy as the gold standard), and its sensitivity for detecting polyps ≥10 mm was 84% to 97% with a specificity of 91% to 99%, showing its potential as an alternative to colonoscopy.[31] However, the variation in completion rate (57%–92%) of capsule colonoscopy examination of the entire colon is a concern. In comparison with colonoscopy, CT colonography (CTC) has a comparable detection rate for AN.[32] However, higher screening participation rates of CTC led to diagnostic yields comparable to colonoscopy in a randomised controlled trial.[33] CTC is now accepted in North Amercian and European countries as an alternative examination to colonoscopy, as the diagnostic test after a positive FIT. In the United States, CTC is used as the primary screening modality or when colonoscopy is incomplete. An important limitation of CTC is the need for excellent bowel preparation. CTC has also had significant false-positive reults due to preparation problems, which reduces the positive predictive values. A study from nationwide multicentre prospective controlled study from Japan compared the performances of CTC and colonoscopy on the same day in a non-screening population with images interpreted by trained gastroenterologists and radiologists. Results showed that CTC is not sufficiently sensitive to detect non-polypoid neoplasms.[34] Most cost-effectiveness studies show that CTC is not cost-effective relative to other screening programmes.[35] In the Asia-Pacific region, many countries have limited access to CTC. Therefore, the working group cannot recommend a stool-based DNA test, capsule colonoscopy or CTC as a first-line screening tool for CRC in the Asia-Pacific region.

Age of screening for CRC

Statement 5. There is a rising trend of young patients (<50 years of age) with CRC in Asia, especially in men and for cancers located in the rectum.

Level of agreement: A=70.8%, B=25.0%, C=4.2%, D=0%, E=0%

Quality of evidence: II-2

Classification of recommendation: A

The rising trend for young patients below the age of 50 years to develop CRC, which was first reported by Siegel *et al*,[36] is no longer a western phenomenon. Studies in Asia have reported a similar trend of early-onset CRC. In a multinational cohort study involving four Asian countries/regions—namely, Taiwan, Korea, Japan and Hong Kong, a similar trend of increasing incidence of early-onset CRC was reported[37] (table 2). The most pronounced change was observed for male rectal cancer, although a similar trend was also evident among women and in the proximal colon. Subsequently, the rising trend of early-onset CRC has also been reported in other Asia-Pacific countries, including Korea,[38] Australia[39] and India.[40] Using a population-based, age-period,

Table 2 Increased incidence of young (<50 year olds) colorectal cancer (CRC) in four countries/regions of the Asia-Pacific region. Average annual percentage change in young-onset CRC (age <50 years) in Hong Kong, Japan, Korea and Taiwan in the period 1995–2014.

	Average annual percentage change (95% CI)			
Country/Region	Colon cancer Male (%)	Colon cancer Female (%)	Rectal cancer Male (%)	Rectal cancer Female (%)
Hong Kong	+0.1 (-1.7 to +1.9)	−0.3 (-3.0 to +2.4)	+2.5* (+1.4 to+3.6)	+0.8* (−0.3 to +1.9)
Japan	+3.3 (-0.1 to +7.8)	+1.8 (-1.9 to +5.6)	+2.0* (+0.4 to+3.6)	+2.1* (+0.8 to+3.4)
Korea	+4.7* (+3.5 to+5.9)	+5.5* (+3.5 to+7.6)	+6.0* (+4.5 to+7.6)	+4.8* (+2.7 to+7.0)
Taiwan	+3.7* (+3.0 to+4.4)	+3.5* (+2.9 to+4.1)	+3.9* (+3.3 to+4.5)	+2.7* (+2.1 to+3.4)

*p<0.05.

cohort study to compare the incidence of CRC in Asia with that in the West, again showed an increasing risk of patients with early-onset CRC in both Western and Asian populations.[41] The rising trend of early-onset CRC is more evident in countries of high economic status, although a global phenomenon.[42] Risk factors associated with early-onset CRC include male gender, inflammatory bowel disease, family history of CRC, obesity, smoking and diabetes.[43 44] The recent rapid increase in incidence may also be related to lifestyle changes, such as intake of simple sugar and sweetened beverage during adolescence.[45] This is an alarming trend that should alert healthcare systems in the Asia-Pacific region to take note and be prepared. It is imperative to look into the effects of lifestyle changes in preventing early-onset CRC.[46] Discussion of the age at which CRC screening should be started is naturally becoming a hot topic of debate.

Statement 6. Lowering the age to start CRC screening to <50 years has not been proved to be cost-effective in Asia.

Level of agreement: A=37.5%, B=45.8%, C=4.2%, D=12.5%, E=0%

Quality of evidence: III

Classification of recommendation: B

With the increasing global incidence of early-onset CRC, the need for lowering the age at which CRC screening is started has been a focus of discussion. The US Multi-Society Task Force on Colorectal Cancer recommended that screening should begin at the age of 50 in people at average risk, except in African Americans in whom limited evidence supported starting screening at age 45.[47] The American Cancer Society was the first to propose lowering the screening age for all adults to 45 years.[48] The most recent USPSTF recommendation and the US Multi-Society Task Force recommend that asymptomatic adults aged 45 years or older at average risk of CRC (ie, no prior diagnosis of CRC or advanced adenoma, inflammatory bowel disease and no family history of known genetic disorder predisposing to CRC) should receive CRC screening.[49 50] This is based on age-period cohort analyses in birth cohorts in the United States and clinical trials of CRC screening that include adults younger than 50 years of age.[51 52] Using screening colonoscopy in a Markov model analysis, a study from the United States showed that a screening programme starting at the age of 45 years is likely to be cost-effective.[53] Furthermore, modelling performed by the Cancer Intervention and Surveillance Modelling Network suggests that starting CRC screening at age 45 years may moderately increase life-years gained and decrease CRC cases and deaths compared with beginning screening at the age of 50.[22] However, modelling studies are based on assumptions that are tailored to specific socioeconomic factors—for example, life-years gained, costs of a screening test and treatment costs for CRC in a specific region or country. The microsimulation modelling analysis, which suggests that CRC screening using a stool test, colonoscopy or CT colonography starting at the age of 45 years provides an efficient balance of colonoscopy burden and life-years, gained may not apply to the Asia-Pacific region. The only study in the Asia-Pacific region comes from Australia, which used a microsimulation model to study the cost-effectiveness of starting screening at 40–45 years and cessation at 79–84 years.[54] The study concluded that starting screening at 45 years could be cost-effective, but would increase colonoscopy demand, jeopardising the benefit of those subject to screening at 50–74 years. The working group chose to recommend that, in view of the vast population in this region, priority should be given to those over the age of 50 years as their risk of CRC is still significantly higher than that of the average-risk population below the age of 50.

There are two caveats to this recommendations. First, the modelling studies are mainly based on colonoscopy-based screening scenarios. It is not clear whether it also applies in a two-tier screening setting using FIT and colonoscopy as a sequential screening tool. Also, given the possibility that young-onset CRC continues to increase in the Asia-Pacific region, further studies in this region using FIT-based screening starting at a younger age may be worthwhile.

Statement 7. Older age per se should not be a reason to exclude healthy individuals from starting screening.

Level of agreement: A=54.2%, B=29.2%, C=8.3%, D=8.3%, E=0%

Quality of evidence: III

Classification of recommendation: C

There are few data addressing the issue of the upper age limit for CRC screening. Most studies did not include people over the age of 75 years. The risk of screening colonoscopy increases with age and therefore, it is not recommended after age 75. The incidence of CRC and adenomas increases with age, but the use of non-invasive screening tests and a sequential screening approach (eg, FIT plus colonoscopy) might mitigate the risk in the older subjects. To discontinue safe, rapid, painless, cost-effective screening with a non-invasive faecal occult blood test, FIT, or stool-based multitarget DNA, or minimally invasive virtual colonoscopy and capsule colonoscopy for individuals older than 75 is neither supported, nor refuted by the literature. This strategy will, however, deny individuals the benefits of prevention and early diagnosis of cancers. CRC or adenomas identified by screening could be addressed with an evidence-based decision, considering the individual's risk tolerance and desires of the patient and family. The Asia-Pacific Working Group considered the ageing population of Asia but also the general improved health status of adults over age of 75, and feels that age alone should not prohibit healthy subjects from screening. Using age alone as a cut-off point for CRC screening is insufficient.[55] In determining whether CRC screening should be offered, patients and clinicians should discuss and make the decision together based on a patient's overall health status, prior screening history and patient's preferences.

Statement 8. Screening can be stopped at age 85 if an individual has a recent negative test as they are unlikely to benefit from further screening

Level of agreement: A=54.1%, B=33.3%, C=4.2%, D=4.2%, E=4.2%

Quality of evidence: III

Classification of recommendation: C

Typical pathogenesis of CRC is an adenomatous polyp slowly increasing in size and accumulating genetic mutations, leading to dysplasia and ultimately, cancer. Most CRCs arise from colorectal adenomas, but 20–30% of cases arise through pathways other than the adenoma-carcinoma sequence. Progression from adenoma to invasive cancer varies and may take an average of 15 years, and in certain instances progress within a shorter time. The American Cancer Society recommended that adults at average risk, in good health with a life expectancy of more than 10 years, should continue CRC screening until the age of 75 years.[48] For individuals aged 76 to 85 years, decisions on CRC screening should be individualised. However, above the age of 85 years, clinicians should discourage individuals from continuing CRC screening. The US Preventive Services Task Force recommendation is to offer screening for adults aged 50 to 75 years. The USPSTF also recommends that clinicians can selectively offer screening for CRC to adults between 76 to 85 years of age.[49] Although the current USPSTF guideline

recommends against screening in patients older than 85 years, other competing health concerns should be considered, as there are no data to show that the benefits of screening cease at any particular age as the only factor. The working group accepts that screening should be stopped at age of 85 years if an individual has had a recent negative test.

Screening for individuals with a family history of CRC or advanced adenoma

Statement 9. Healthcare providers should always explore the family history of their patients with CRC for the possibility of hereditary CRC syndromes.

Level of agreement: A=91.7%, B=8.3%, C=0%, D=0%, E=0%

Quality of evidence: III

Classification of recommendation: A

Approximately 30% of patients with CRC have a family history of this disease.[56] Some families with a history of CRC carry genetic variants that cause CRC with high or moderate penetrance, but these syndromes account for only around 5% of CRC cases. Recent germline cancer testings including more extensive panels of moderate risk genes suggest a higher association of 12–15%.[57] Most families with a history of CRC and/or adenomas do not carry specific genetic variants associated with cancer syndromes. Family history of CRC includes three categories: (1) those with recognised familial genetic syndromes; (2) those who have first-degree relatives (FDRs) with CRC; their age at diagnosis may be young (<50 years) or older (>50 years) and (3) those having FDRs with documented advanced adenomas or SSLs. A positive family history increases the risk of developing CRC. Furthermore, those with a positive family history tend to develop CRC at a younger age.[58]

Many national and international guidelines on CRC screening include recommendations on screening of individuals with family history of CRC.[47 59] (table 3). Where carriers of penetrant genes are identified (eg, Lynch syndrome, familial adenomatous polyposis, MUTYH- associated polyposis), syndrome-specific guidelines cover the age of onset and frequency of colonoscopic screening and address screening and risk-reducing strategies for non-colorectal syndrome specific cancers. For Lynch syndrome, MMR gene specific guidelines are now promulgated, recognising gender- and gene-specific differences in penetrance. Although the benefits of colonoscopy are currently being questioned owing to the recognition of a non-polypoid pathway to CRC in Lynch syndrome and the possibility of overdiagnosis,[60] guidelines are still unanimous on the need for high-quality colonoscopy.

In Asia, very few genetic studies have been conducted specifically for CRC. Possibly a substantial number of high-risk asymptomatic individuals with family history of advanced neoplasia and CRC miss the opportunity of early diagnosis and treatment. Healthcare workers, including gastroenterologists, surgeons, family physicians, and nurses, should explore the family history of CRC in their patients. Studies have shown that both patient's awareness of the importance of a family history of CRC as well as the recommendation of a family physician to undergo CRC screening are the two most powerful incentives for individuals at average risk to be enrolled in a screening programme.[61] Guidelines for surveillance of people with a family history of CRC or advanced adenomas are very much needed.

Statement 10. Subjects having two FDRs with CRC or advanced adenoma diagnosed at any age should start screening at 10 years before the age at diagnosis of the youngest affected FDR or at age 40 (whichever is earlier) with colonoscopy every 5 years

Table 3 Comparison of guidelines of professional societies on screening of individuals with family history of colorectal cancer or advanced adenoma

Organisation	Family history	Recommendations
National Comprehensive Cancer Network[162 163]	1 FDR with CRC or endometrial cancer diagnosed at <50 years, or with synchronous or metachromous Lynch syndrome-related cancers Other 1 FDR with CRC	Genetic evaluation Start colonoscopy at age 40 years, or 10 years before the earliest diagnosis of CRC in the family, with surveillance at least every 5 years
US Multi-Society Task Force on Colorectal Cancer (ACG, AGA, ASGE)[47]	1 FDR with CRC or AA at <60 years or ≥2 FDRs at any age 1 FDR with CRC or AA at ≥60 years	Start colonoscopy at 40 years or 10 years before earliest diagnosis (whichever is earlier), repeat every 5 years Start screening at 40 years (any modality), repeat as average risk
American College of Gastroenterology[164]	1 FDR with CRC or AA at <60 years or ≥2 FDRs at any age 1 FDR with CRC or AA at ≥60 years For individuals with higher familial CRC burden	Start colonoscopy at 40 years or 10 years before earliest diagnosis, repeat every 5 years Start colonoscopy at 40 years, or 10 years before earliest diagnosis, then resume screening according to average-risk recommendations Genetic evaluation
Canadian Association of Gastroenterology[59]	1 FDR with CRC or AA >2 FDRs with CRC or AA	Start colonoscopy at 40–50 years or 10 years before earliest diagnosis, repeat every 5–10 years Start colonoscopy at 40 years or 10 years before earliest diagnosis, repeat every 5 years
Cancer Council Australia[165]	1 FDR with CRC at <55 years, or 2 FDRs at any age, or 1 FDR and at least 2 SDRs with CRC at any age ≥3 FDRs or SDRs with CRC, with at least one at <55 years; or ≥2FDR with CRC at any age	iFOBT every 2 years from 40 to 49 years. Start colonoscopy at 50 years, repeat every 5 years until 74 years, iFOBT every 2 years from 35 to 44 years. Start colonoscopy at 45 years, repeat every 5 years until 74 years
European Society of Gastrointestinal Endoscopy (ESGE)[166]	1 FDR with CRC at <50 years, or ≥2 FDRs with CRC at any age	Start colonoscopy at 40 years, repeat every 5 years
British Society of Gastroenterology (BSG) and Association of Coloproctology of Great Britain and Ireland (ACPGBI)[167]	1 FDR with CRC <50 years, or ≥2 FDRs with CRC at any age 3 FDR across >1 generation	Colonoscopy at 55 years, then repeat as usual postpolypectomy surveillance guideline Start colonoscopy at 40 years, then repeat every 5 years until 75 years

AA, advanced adenoma; ACG, American College of Gastroenterology; AGA, American Gastroenterological Association; ASGE, American Society for Gastrointestinal Endoscopy; CRC, colorectal cancer; FDR, first-degree relative; iFOBT, immunochemical faecal occult blood test; SDR, second-degree relative.

Level of agreement: A=45.8%, B=41.7%, C=12.5%, D=0%, E=0%
Quality of evidence: II-2
Classification of recommendation: A

Statement 11. Subjects having one FDR with CRC or advanced adenoma diagnosed at age <60 years should start screening at 10 years before the age at diagnosis of the youngest affected FDR or age 40 (whichever is earlier) with colonoscopy every 5 years

Level of agreement: A=41.6%, B=41.7%, C=12.5%, D=0%, E=4.2%
Quality of evidence: II-2
Classification of recommendation: A

Statement 12. Subjects having one FDR with CRC diagnosed at age ≥60 years should start screening at the age of 40 with the same test strategy and intervals as for average-risk individuals.

Level of agreement: A=50%, B=45.8%, C=4.2%, D=0%, E=0%
Quality of evidence: II-2
Classification of recommendation: A

Evidence suggests that having a FDR with advanced adenomas carries a similar increased risk for CRC as having an FDR with CRC. In a blinded, cross-sectional study, 200 asymptomatic siblings of patients with advanced adenomas (adenomas >10 mm, high-grade dysplasia and villous structure) and 400 age- and sex-matched siblings of subjects with normal findings from colonoscopies and no family history of CRC underwent colonoscopy at two hospitals in Hong Kong.[62] The prevalence of advanced adenomas was 11.5% among subjects with one FDR diagnosed with CRC and 2.5% among subjects without such a family history (matched ORs=6.05, 95% CI 2.74 to 13.36). On the other hand, non-advanced adenoma do not carry the same risk of CRC to a FDR.[63] Therefore, FDRs of those with CRC or advanced adenoma should start screening for CRC at an earlier age.

Starting age for screening depends on the age of diagnosis of the proband in the family and the number of FDRs with CRC or advanced adenoma. In a Utah population-based study, FDRs of patients with CRC had a 1.8-fold increased risk of CRC (hazard rate ratio 1.79, 95% CI 1.59 to 2.03), as did second-degree relatives (hazard rate ratio 1.32, 95% CI 1.19 to 1.47).[64] It should be pointed out that the risk of relatives in the Utah study was not age related— that is, the risk, though elevated, was not at an early age if the proband was young. This risk was greater for FDRs when index patients developed CRC at less than 60 years of age than in those older than 60. Individuals who have two or more FDRs diagnosed with CRC or advanced adenoma also have a higher risk than those with only one FDR. Thus, the number of FDRs involved and the age of onset of disease are the two crucial factors to start screening earlier than at age 50.

Although there is clear evidence of an increased risk in this group, data are limited on its magnitude relating to the age of CRC onset of the proband. Therefore, there is a great variety across different guidelines from various national and professional societies (table 3). The definition of who should undergo intensified colonoscopy surveillance instead of average-risk screening also varies widely (table 3). Nevertheless, the consensus accepted starting screening at an age 10 years earlier than the earliest age of onset in the family as the overall risk at an earlier age in these circumstances surpasses the risk a decade or two later in average-risk populations. Thus, the number of FDRs involved and the age of onset of disease are the two crucial factors to start screening earlier than at age 50.

The Asia-Pacific Working Group considers that lacking large-scale cohort studies in the region to make alternative recommendations, we should follow the international guidelines—that is, using the number of FDRs and the age of diagnosis of CRC or advanced adenoma in the FDRs as risk stratification criteria. Subjects having two FDRs with CRC or advanced adenoma, diagnosed at any age, are recommended to start screening at 10 years before the age at diagnosis of the youngest affected FDR or at age 40 (whichever is earlier) with colonoscopy every 5 years. Subjects having one FDR with CRC or advanced adenoma diagnosed at age <60 years should start screening at 10 years before the age at diagnosis of the youngest affected FDR or age 40 (whichever is earlier) with colonoscopy. These two are considered the groups at highest risk. Subjects having one FDR with CRC diagnosed at age ≥60 years are considered lower risk and should start screening at the age of 40 with the same test strategy and intervals as for average-risk individuals—that is, colonoscopy or FIT is acceptable.

Surveillance for individuals with one or more adenomas

Statement 13. Surveillance colonoscopies represent a heavy burden to the healthcare system and to individuals and the optimal method for surveillance and intervals should be studied in Asia.

Level of agreement: A=87.5%, B=12.5%, C=0%, D=0%, E=0%
Quality of evidence: III
Classification of recommendation: A

Individuals with adenomas of different endoscopic and pathological features at colonoscopy carry different risks of developing metachronous advanced neoplasm. Surveillance colonoscopy provides these individuals with additional protection for incident CRC. Occurrence of CRC after colonoscopy could be secondary to inadequate quality of colonoscopy or be due to specific tumour biological behaviour (eg, MSI tumours). Surveillance colonoscopy provides an opportunity to detect overlooked or newly developed neoplasms before they become invasive or advanced-stage CRC.[65] The frequency of undergoing surveillance colonoscopy should be tailored according to the individual risk of developing metachronous CRC, thereby making the most efficient use of the constrained resources such as colonoscopy capacity. A remarkable degree of overuse or underuse of surveillance colonoscopy exists and needs to be optimised based on the current evidence and the clinical guidelines.[66–68] In addition, surveillance guidelines in the Asia-Pacific also require revision based on the most updated data on colonoscopy use and outcomes.[69]

A large body of literature has shown that those with three or more adenomas or adenomas with advanced histology are at higher risk of developing metachronous advanced neoplasms after colonoscopy.[70–73] These tumours are more often flat, proximally located and likely to be missed or incompletely resected.[74] More recent studies show that individuals with low-risk adenomas (defined as one or two tubular adenomas <10 mm) followed up for 7–15 years have an incidence of CRC comparable to those with no adenoma, and significantly lower than for those with high-risk adenoma (defined as three or more adenomas or adenomas with villous or high-grade dysplasia histology or size ≥10 mm).[75–78] The European guideline (ESGE) recommends that five low-risk adenomas should be considered as high-risk instead of three, which further reduce the frequency of surveillance colonoscopy.[67] Surveillance intervals could be tailored according to the risk levels to avoid unnecessary colonoscopies.

High-quality baseline colonoscopy (complete caecal intubation, adequate bowel preparation, appropriate withdrawal time, adequate adenoma detection rates, complete polyp resection and low adverse events and complications)[79] is the premise of recommending different surveillance intervals based on colonoscopic findings. Adenoma detection rates (ADRs), one of the most important metrics of colonoscopy quality, have been shown to be inversely associated with the risk of incident CRC after baseline colonoscopy, incident advanced-stage CRC and even CRC death in a colonoscopist's practice.[80–82] Incomplete endoscopic resection of neoplasms detected at baseline colonoscopy or new-onset neoplasms are other causes for the development of postcolonoscopy CRC.[83–86]

Statement 14. After removal of high-risk adenomas, surveillance colonoscopy should be done at 3 years

Level of agreement: A=58.3%, B=25%, C=8.3%, D=4.2%, E=4.2%

Quality of evidence: II-2

Classification of recommendation: A

It is well studied that individuals with one or more high-risk adenomas (high-grade dysplasia (HGD) or villous component in histology, 10 mm or larger in size, and three or more in number) are at a higher risk of developing metachronous advanced adenomas or CRC after colonoscopy.[87] Approximately 4% to 10% of the colonoscopic screening participants belong to this category and currently, 3-year surveillance is recommended for this subgroup.[76–78 88] Such a surveillance interval is based on the natural history of the colorectal neoplasia, with the average dwelling time of 3 to 4 years for an advanced adenoma to progress to invasive cancer, and the results of previous randomised trials, including the result from the National Polyp Study.[89–92] A recent study by Wieszczy *et al* demonstrated that those subjects with an advanced adenoma ≥2 cm or HGD are at higher risk of metachronous CRC than those with other adenomas. Such data may help to further reduce the proportion of subjects who require intensive surveillance by threefold.[77] A prospective multicentre trial was conducted in 11 Japanese institutions, randomising patients to receive follow-up colonoscopy at 1 and 3 years or at 3 years only after detecting advanced neoplasia at baseline colonoscopy.[92] In this study of 3926 patients with a mean age of 57 years, no difference in the detection of adenomas >10 mm, HGD or cancers was found between the two strategies. Based on this study, a 3-year interval is considered appropriate for Asian individuals. The recommended surveillance interval by the Asia-Pacific Working Group after the removal of high-risk adenomas is generally in line with the 3-year surveillance interval recommendations by the Japanese and Korean guidelines.[93 94]

Statement 15. Surveillance for low-risk adenomas (<3) should be done by colonoscopy and/or FIT at intervals as used for average-risk individuals.

Level of agreement: A=70.8%, B=29.2%, C=0%, D=0%, E=0%

Quality of evidence: II-2

Classification of recommendation: B

Five-year surveillance intervals have been recommended by major guidelines for individuals with low-risk adenomas. Based on recent studies showing a similar risk of incident CRC after colonoscopy in this low-risk group compared with a negative screening colonoscopy in the average-risk population,[76–78] current recommendations suggest prolonging the surveillance interval to 7–10 years, or even the same as the interval for those individuals without adenomas (ie, 10-year interval). It should be noted that such a long surveillance interval is based on a high-quality baseline colonoscopy, and inadequate quality may lead to the occurrence of postcolonoscopy colorectal cancer (PCCRC).

In some FIT screening programmes or guidelines, regular FIT screening is recommended after negative colonoscopy.[72] A recent study from Taiwan showed that subsequent FIT after a negative colonoscopy had positivity rate of 11.3%, and for every 12 colonoscopies one CRC was detected. This was associated with a reduction of incident CRC by 53% and ameliorated the risk of PCCRC caused by inadequate colonoscopy quality.[95] Therefore, regular FIT in between colonoscopy seems to be a prudent strategy and may add further reassurance about the prevention and early diagnosis of CRC. Using serial FIT in the intermediate-risk group (three to four small adenomas or on e≥10 mm) has also been discussed. Cross *et al* reported that annual FIT after colonoscopy in such risk groups reduced the volume of colonoscopy by 71% but missed 30–40% of CRCs and 40–70% of advanced adenomas, which would have been detected by colonoscopic surveillance.[96] In Asia, where the volume of colonoscopy is high and resources are limited, FIT as an alternative to colonoscopy in surveillance is definitely appealing. The use of a FIT for surveillance instead of colonoscopy-based surveillance needs further evaluation. For the current recommendation, the working group proposes surveillance for low-risk adenomas (<3), which should be performed by colonoscopy and/or FIT at intervals as used for average-risk individuals. Further specific recommendations on FIT surveillance require more data.

Screening and surveillance for sessile serrated lesions

Statement 16. As sessile serrated lesions are premalignant lesions, they should be detected and removed.

Level of agreement: A=83.3%, B=16.7%, C=0%, D=0%, E=0%

Quality of evidence: II-3

Classification of recommendation: B

For a long period of time, colorectal adenomas were thought to be the only precursor lesions for CRC. However, recent studies show that, despite a high adenoma detection rate, PCCRCs still occur, which are more likely to be related to serrated lesions mostly located in the proximal colon.[97] New knowledge in histopathology has now clarified that serrated lesions are a heterogeneous group that includes both premalignant lesions and clinically insignificant small hyperplastic polyps. Almost a decade after publishing the fourth edition of WHO classification of tumours of the digestive system, the fifth edition WHO classification in 2019 has adopted the new term SSLs, which refers to serrated premalignant lesions that were formerly called sessile serrated adenomas (SSAs) or sessile serrated polyps (SSPs)[98 99] (table 4). The change of terminology from adenoma/polyp to

Table 4 Changes in WHO Classification of serrated colorectal lesions

WHO classification of serrated colorectal lesions, 4th edition 2010	WHO classification of serrated colorectal lesions, 5th edition 2019
Hyperplastic polyp (HP) ► Microvesicular HP (MVHP) ► Goblet cell-rich HP (GCHP) ► Mucin-poor HP (MPHP)	Hyperplastic polyp (HP) ► Microvesicular HP (MVHP) ► Goblet cell-rich HP (GCHP)
Sessile serrated adenoma/polyp (SSA/P) with or without dysplasia	Sessile serrated lesions (SSLs) condition serrated lesions with dysplasia (SSLD)
Traditional serrated adenoma (TSA)	Traditional serrated adenoma (TSA)
	Serrated adenoma, unclassified

lesions highlighted that these serrated lesions might not show morphologic dysplasia or a polypoid structure, and hence calling them adenomas or polyps might be inappropriate and confusing. Hyperplastic polyps (HPs) are still considered to be benign lesions with no risk of malignant transformation. The subtype mucin-poor hyperplastic polyp has been deleted leaving only microvascular and goblet cell-rich subtypes.[100] Yet, the malignant potential of SSL is confirmed and may contribute significantly to PCCRCs.[101] Studies comparing prevalent CRC discovered at screening with incident CRC discovered during surveillance, found that the incident CRC tend to be more proximal and have characteristics compatible with SSL with CPG island methylation and MSI-H.[97 102] Traditional serrated adenomas (TSAs) are rare but clearly also a precursor of malignancy. Both TSAs and SSLs may progress via the serrated pathway to CRC.[100]

The diagnostic criteria for SSLs have also been clarified. Instead of 'two or three' distorted serrated crypts in a polyp in the old classification,[98] the updated 2019 WHO classification[99] specified that as a minimum requirement, one distorted serrated crypt is sufficient for the diagnosis of SSL. The definition of an architecturally distorted serrated crypt is now specified as showing at least one of the following histologic features: (1) horizontal growth along the muscularis mucosa (L-shaped or inverted T-shaped crypt); (2) dilatation of the crypt base (basal one-third of the crypt); (3) serrations extending into the crypt base; (4) asymmetrical proliferation of the crypts. Although SSLs have been regarded as a premalignant lesion for CRC, only a small proportion of them show morphologic dysplasia (SSL-D). This suggests that once dysplasia develops within an SSL, the progression to cancer is accelerated.[103] There are thought to be three types of dysplastic changes in SSLs[104]:

1. Intestinal dysplasia: similar to the dysplasia in adenomas, typified by elongated, pseudostratified, hyperchromatic nuclei with basal cytoplasm;
2. Serrated dysplasia: characterised by cuboidal cells with eosinophilic cytoplasm, increased nuclear prominence, atypia and mitosis, and its presence can be considered to represent progression to a TSA.
3. Minimal deviation dysplasia: with limited changes compared with SSL, but with the characteristic loss of MLH1.

Most cases of SSL-D have an undefined pattern of dysplasia, with the majority showing a loss of MLH1 expression. However, the clinical significance of these histologic differences in dysplasia is not fully known. Is the finding of SSL significant? In a matched cohort study through prospective record linkage in Sweden among patients aged 18 years or older with a first diagnosis of colorectal polyps in the nationwide gastrointestinal ESPRESSO histopathology cohort (1993–2016), CRC mortality rates of patients diagnosed with tubulovillous adenomas, villous adenomas as well as SSLs were higher than those in controls.[105] Recently it has been shown in a FIT-positive screening population that, just like ADR and alongside the ADR, the proximal serrated polyp detection rate is inversely correlated with the incidence of interval postcolonoscopy CRC.[100] Therefore, SSLs are indeed associated with CRC. With these clarifications in criteria of diagnosis and updated classification of SSLs, the Asia-Pacific Working Group feels that it is mandatory for gastroenterologists to detect and remove all SSLs to prevent CRC.

Statement 17. Sessile serrated lesions are likely to be underdiagnosed in Asia and should be carefully looked for during colonoscopy, especially in the right colon.

Level of agreement: A=79.2%, B=20.8%, C=0%, D=0%, E=0%

Quality of evidence: II-2

Classification of recommendation: B

The serrated pathway is now believed to account for 30% of all cases of CRC but, as detection rates vary widely among endoscopists and pathologists, there is uncertainty about the prevalence of these lesions in our locality.[106] Prevalence varies with study location, diagnostic criteria and examination quality. Relatively little is known about the epidemiology of these lesions (prevalence, location, family history) and risk of malignant transformation (timing, associated factors). With the changes in diagnostic criteria (see above) and awareness of the condition, new prevalence studies should be performed. A systematic review of 74 publications on SSPs was done, after screening 4462 papers extracted from the literature.[106] Before 2010, few reports on the prevalence of SSPs were published, but this number increased dramatically after release of the WHO 2010 classification. Reported prevalence of SSPs was 3.9% in Europe, 5.1% in the USA and 2.6% in Asia. Australia reported an exceptionally high prevalence of 10.5%[106] because of one statistical outlier.[103] The US and European studies also showed substantial heterogeneity in estimates of SSL prevalence, probably due to varying detection rates between endoscopists and different baseline population risk. Under the old terminology, the prevalence of clinically relevant serrated polyps in US and European studies varied from 1.8% for large hyperplastic polyps to 3.3% for serrated adenomas to 8.5% for proximal serrated polyps.[103]

In Asia, only a few reports on SSLs have been published. Excluding one extreme statistical outlier, the mean prevalence of SSLs is much lower in Asia than in the United States—namely, less than 1%.[106] On the other hand, a study from Japan recruiting 5218 asymptomatic subjects for CRC screening reported detection rates of serrated lesions of 23.3% and of right-sided serrated lesions of 7.6%, respectively.[107] In that study, high-quality video endoscopes with narrow-band imaging and magnification were used with 0.4% indigo carmine dye to enhance the detection of flat lesions. On the other hand, in another CRC screening study from Hong Kong recruiting 6011 subjects, 486 (8.1%) subjects were reported to have HPs and only 85 (1.4%) SSLs.[108] A retrospective study from China reported SSLs in only 0.7% among symptomatic and asymptomatic individuals who underwent colonoscopy.[109] In Australia, the prevalence of SSLs in Chinese (2%) was lower than in Caucasian (7%) subjects.[110] Taken together, SSLs are likely to be underdiagnosed in Asia and should be carefully looked for during colonoscopy, especially in the right side of the colon. Training for endoscopists and pathologists to identify SSLs will probably increase detection rates and improve the prevalence of estimates of these lesions.

Statement 18. Detection of sessile serrated lesions may be enhanced by improving awareness of the importance of these lesions and training in their detection.

Level of agreement: A=87.5%, B=12.5%, C=0%, D=0%, E=0%

Quality of evidence: I

Classification of recommendation: A

The high variability between studies for the prevalence of SSLs, is at least partly explained by varying detection rates of serrated lesions between endoscopists, as this rate appears highly operator dependent. In a study involving 15 experienced gastroenterologists at two academic endoscopy units between 2000 and 2009, during which a total of 11 049 polyps were detected in 6681 colonoscopies, the detection rate of serrated polyps in the proximal colon varied from 1% to 18%.[111] Two recently published meta-analyses also reported the detection rates of SSLs. Results from 16 studies reporting SSL detection rates and 12 studies reporting proximal serrated polyp detection

rates were pooled.[112] Among 280 370 average-risk individuals undergoing screening colonoscopy, SSLs were detected in 2.5% and proximal serrated polyps in 10%. Heterogeneity and publication bias was noted in both this review and meta-analysis. In a similar meta-analysis by another group, including 17 studies and 129 000 colonoscopies, SSLs were detected in 2.5%.[113] Detection of SSLs was higher among Caucasians (2.9%) than among Asians (0.7%). Subgroup analysis showed that SSLs, with an overall detection rate of 19%, were more common in men (22%) than in women (14%), and more common in Caucasians (25.9%) than in Asians (14.6%). Increasing both endoscopist and pathologist awareness, as well as interdisciplinary communication is one way to increase detection of SSLs.[114 115]

SSLs are sessile or flat lesions with an average size of 5–7 mm. They are usually larger than HPs. They may exhibit distinct endoscopic features, such as overlying mucus cap, cloud-like surface, ring of debris or stool around the lesion and obscured mucosal vasculature.[116 117] During narrow-band imaging (NBI) endoscopy, they have a cloud-like appearance, irregular shape and dark spots inside the crypts. Better bowel preparation, longer withdrawal time and careful examination of the right colon (with repeated anterograde examination or retroflexion in the caecum) improved detection of SSLs.[104] Chromoendoscopy such as NBI may marginally improve the detection of SSLs but is currently not recommended as mandatory practice as clear scientific evidence is lacking.

Dutch Workgroup serrAted polypS and Polyposis (WASP) developed a classification system for optical diagnosis of adenoma, HP and SSA/P[118] by combining the NBI International Colorectal Endoscopic classification (1. brown colour, 2. brown vessel, 3. oval tubular or branched surface patter) with the WASP criteria for differentiation of SSLs (1. clouded surface, 2. indistinct borders, 3. irregular shape, 4. dark spots inside the crypts) in a stepwise approach. After training, the accuracy of optical diagnosis of SSL improved.[119] Just a brief and simple educational classroom intervention to trained and untrained endoscopists significantly improved the detection of serrated polyps and the improvement was long term.[119] The working group recommends setting up proper training opportunities for endoscopists to optimise SSL detection rates.

Statement 19. After complete removal of one SSL >10 mm or SSL with cytological dysplasia, colonoscopy should be recommended at a 3-year interval.

Level of agreement: A=54.1%, B=37.5%, C=4.2%, D=4.2%, E=0%

Quality of evidence: III

Classification of recommendation: B

PCCRCs are defined as cancers diagnosed after a colonoscopy in which no cancer was found before the next due surveillance procedure. They might arise from a specific biologic type of CRC, missed cancer as well as missed or incompletely resected benign lesions.[120] Studies have shown that PCCRCs occur in 1–9% of cases and are more commonly found in the right colon.[121 122] Molecular profiling studies have found that PCCRCs more often display the molecular features CIMP (OR 2.19) and MSI (OR 2.1), indicating that they often followed the serrated pathway and more often arise from serrated polyps.[19 123]

There is convincing evidence that patients with serrated polyps have an increased risk of CRC. A study using the Danish database including 272 342 individuals who underwent colonoscopies showed that having SSAs/Ps at baseline colonoscopy increased the risk of metachronous CRC about threefold.[124] In the New Hamsphire Colonoscopy Registry study, 65 patients with large serrated polyps (HP, SSP or TSA) underwent two colonoscopies with a median time surveillance of 4.9 years, of whom 3.1% had a high-risk adenoma.[125] SSA/Ps with cytological dysplasia have an even higher risk of CRC of 4.76-fold. SSLs with a nodule/protusion and those larger than 10 mm in size are more likely to have cytological dysplasia.[126] However, all SSLs should be removed and all those individuals carefully surveilled after removal. Size and multiplicity are the two most important risk factors for metachronous malignancy.[127]

During colonoscopy, SSLs can be removed (piecemeal) by cold snare polypectomy, (piecemeal) by endoscopic mucosal resection and under water endoscopic mucosal resection. A study in four Australian centres showed that, irrespective of the method of removal, around 4% of SLLs recur within 18 months.[128] A similar recurrence rate of 5% within 18 months was reported in Japan.[129] Another study reported a 3.6% recurrence within 18 months.[130] A meta-analysis pooling nine studies with 34 084 participants confirmed that individuals with serrated polyps have a twofold risk of having advanced adenoma.[131] The presence of proximal serrated polyps and large serrated polyps further increases the risk of advanced adenomas by 2.7-fold and 4.1-fold, respectively.

Based on this evidence, the US Multi-Society Task Force on Colorectal Cancer issued a series of recommendations for follow-up after colonoscopy and polypectomy of SSLs.[47] For patients with low-risk SSLs (one or two lesions, <10 mm in size and completely removed), repeat colonoscopy in 5–10 years is recommended. For patients with medium-risk SSLs (three or four SSLs,<10 mm in size) repeat colonoscopy is recommended in 3–5 years. For those with 5–10 SSLs, <10 mm in size or any SSL ≥10 mm, the US Multi-Society Task Force recommends repeat colonoscopy in 3 years. As the task force mentioned in their guidelines, these are weak recommendations based on very low-quality of evidence (mostly cohort studies, with relatively small sample size and limited follow-up). On the other hand, the ESGE guideline update in 2020 suggests that any serrated polyps <10 mm without dysplasia do not require endoscopic surveillance and those individuals should return to routine screening.[67] An individual with any SSL ≥10 mm or SSL with dysplasia should receive surveillance colonoscopy after 3 years. The European guideline graded the evidence as moderate quality and classified the recommendation as strong. There is obvious discrepancy between the two major international guidelines. The Asia-Pacific Working Group feels that it is important to emphasise the importance of high-quality examination, excellent bowel preparation and complete removal of SSLs. Under these circumstances, after complete removal of any SSL >10 mm or with cytological dysplasia, colonoscopy should be recommended at a 3-year interval.

Quality assurance of CRC screening programme

Statement 20. The quality of a FIT programme depends on adherence to initial testing and follow-up of the testing results.

Level of agreement: A=79.2%, B=20.8%, C=0%, D=0%, E=0%

Quality of evidence: II-2

Classification of recommendation: A

CRC screening is not a single diagnostic test but a programme with steps requiring adherence and follow-up of the testing results. Participation in FIT screening and compliance with diagnostic examinations after a positive FIT is crucial to secure the effectiveness of a FIT screening programme, and these parameters should be regularly audited.[132 133] Any barrier to FIT screening and subsequent diagnostic examinations should be

removed. Education of the public and reminders of serial FITs should be emphasised.

Currently, participation rates of the screening population in the Asia-Pacific region are 43.5% in Australia (2019), 47.8% (men) and 40.9% (women) in Japan (2019), 58.4% (ever-screening rate) in Korea and 38% in Taiwan (2019).[16 134–136] The rate of diagnostic colonoscopy after a positive FIT is approximately 70% in Japan and Taiwan, 62% in Australia, and 61.3% in Korea.[16 134 137] Studies from Taiwan showed that CRC death was 64% higher in FIT-positive individuals who were not compliant with the diagnostic colonoscopy.[138] Both FIT screening participation and colonoscopy rate in the Asia-Pacific region were much lower than those in the FIT-based Dutch and UK programmes, which have a higher than 60% FIT participation and around 85% colonoscopy adherence rates.[139] In the Asia-Pacific region, CRC screening programmes very much rely on a FIT. Adherence to a programme of regular FITs and follow-up actions when testing positive should be strengthened.

Statement 21. Timely colonoscopy (<3 months) after a positive FIT in a CRC screening programme should be offered and monitored.

Level of agreement: A=75%, B=25%, C=0%, D=0%, E=0%

Quality of evidence: II-2

Classification of recommendation: B

Subjects with a positive non-invasive screening test are at higher risk of having advanced adenoma and CRC. Taking FIT as an example, subjects with positive FIT are at a four- to five-fold risk of having an advanced adenoma and 20- to 30-fold risk of CRC compared with the general population of screening age, and also have a higher likelihood of harbouring large and/or synchronous neoplasias.[140] The risk of CRC and advanced adenoma is dependent on the test-positivity threshold.[141] A study from China using the Asia-Pacific CRC Screening Score combined with a FIT showed improvement in positivity of colonoscopy.[142] Delay in diagnostic colonoscopy after a positive FIT might lead to higher CRC incidence, more advanced stages of CRC, and more CRC deaths.[143] The study by Lee *et al* demonstrated that the risk for CRC was 31% higher when colonoscopy was delayed by more than 6 months, and 109% higher for advanced-stage disease. Those risks continuously increased when colonoscopy was delayed by more than 12 months: for any CRC a 117% increase, and for advanced-stage CRC 184%.[144] Similar results were also reported from other programmes.[143 145 146] It is worthwhile mentioning that non-compliance or delay in colonoscopy became remarkable after the outbreak of the COVID-19 pandemic, and these screening and colonoscopy backlogs should aggressively be dealt with to minimise the negative impact on the effectiveness of entire screening programmes and the psychological distress of the FIT-positive subjects.[147–149] In view of this, the working group recommends that timely colonoscopy after a positive non-invasive screening test in a CRC screening programme should be offered, preferably within 3 months after a positive FIT. The follow-up of a FIT should be monitored and used as a performance indicator of the CRC screening programme.

Statement 22. Barriers to screening test uptake and compliance with colonoscopy in CRC screening should be investigated and those present removed.

Level of agreement: A=91.7%, B=8.3%, C=0%, D=0%, E=0%

Quality of evidence: III

Classification of recommendation: B

It is important to identify facilitators and remove barriers to participation in CRC screening. A meta-analysis demonstrated that awareness of CRC affects views, attitudes and motivation for CRC screening.[150] Factors enhancing awareness included public education to address misconceptions, enlisting the efforts of primary care physicians to recommend screening, and enhancing the influence of friends and family of patients with CRC. Significant barriers to participation included barriers to access, logistical challenges to attending screening and cultural beliefs.[150] Previous studies in the Asia-Pacific countries/region showed that insufficient public awareness was the main cause of the low rates of screening participation. Recommendation by a physician is the strongest factor motivating people to go for screening in the opportunistic setting.[151 152] As for the compliance to follow-up procedures after a positive FIT, lack of comprehension about the test was the strongest predictor of non-adherence, whereas higher socioeconomic status, higher perceived threat, higher cues for action, lower perceived barriers, and higher health behaviour scores were identified as factors associated with a greater willingness to undergo diagnostic colonoscopy after a positive FIT.[153 154] Individual programmes should increase efforts to remove or overcome those barriers to follow-up colonoscopy, thereby maximising the effectiveness of the screening.

Statement 23. Quality control of a FIT and colonoscopy is mandatory for CRC screening programmes, and benchmarks should be determined.

Level of agreement: A=91.7%, B=8.3%, C=0%, D=0%, E=0%

Quality of evidence: II-2

Classification of recommendation: A

Assurance of the quality of a FIT and colonoscopy is crucial in securing effectiveness of the entire screening programme. A false sense of security can be generated by a screening programme not properly run.

The performance of a FIT may be affected by the design of the collection tube, the buffer that is used, the ambient temperature, the duration from stool collection to assaying in the laboratory and different FIT brands.[155] In a study from Taiwan, two brands of FITs used in the screening programme were compared and it was demonstrated that one FIT outperformed the other, with better detection rates for CRC and lower rates of interval CRC after a negative FIT, even using the same cut-off point.[156] In the Asia-Pacific, some programmes use more than one brand of FIT kits. In Korea, it was demonstrated that positivity rate of a FIT was lower in the summer, with subsequently higher risk of interval CRC in this season.[157] Regular monitoring of the performance of a FIT at a regional and national level using short-term indicators (positivity rate, detection rate for CRC) and long-term indicators (interval CRC after a negative FIT) is important.

Inadequate quality of colonoscopy is related to PCCRC. It is well-known that the quality of colonoscopy varies between individual programmes, institutes and endoscopists in the rate of complete colonoscopy, detection rates of neoplastic lesions and polypectomy techniques.[121 158] The ADR, one of the most important quality metrics of colonoscopy, is associated with the risk of PCCRC, advanced- stage CRC and CRC death.[80–82] High ADR has been proved to be associated with a reduced risk of interval CRC and CRC death.[159] A multinational study in the Asia-Pacific region has revealed that overall ADR, male-specific ADR, female-specific ADR and overall advanced ADR were 1.93-, 2-, 1.94- and 8.2-fold higher, respectively in the FIT-positive colonoscopy cohort in comparison with the direct screening colonoscopy cohort.[160] Therefore, ADR benchmarks should be higher among FIT-positive individuals. Monitoring and auditing, together with continued training programmes and accreditation systems, will be indispensable to sustain

high-quality colonoscopy screening for CRC.[161] The Asia-Pacific Working Group feels strongly that quality control of colonoscopy and FIT is mandatory for CRC screening programmes, and benchmarks should be determined.

SUMMARY

There are several important updates and revisions in this Asia-Pacific Consensus which should be highlighted. First, despite the trends of younger age of onset of CRC, moving the age of screening to below 50 years of age is not yet recommended for Asia-Pacific countries/regions. This is based on the fact that by far the greatest CRC burden is still in the group aged 50–75 years. Also because of the large burden of screening and surveillance due to the huge population in the Asia-Pacific region, a stratified offering of screening tests based on any form of risk assessment (without preference for one specific scoring system) is strongly recommended. More emphasis should be put on identifying those individuals with a family history of CRC and advanced adenomas in the Asia-Pacific region and following this up with colonoscopy for these individuals. Starting screening at a younger age is recommended among family members of patients with CRC and advanced adenoma. Surveillance for individuals with adenoma should be tailored to the risk of adenomas that were identified and removed at colonoscopy to reduce unnecessary repeated colonoscopic examinations. On the other hand, SSLs have been largely ignored in the Asia-Pacific region and could be an important cause of PCCRC. These lesions should be carefully studied in Asia and colonoscopists should be trained to avoid missing these lesions during endoscopy. Research should focus on surveillance of adenomas and SSLs in the Asia-Pacific region. Finally, quality control of colonoscopy and stool tests is mandatory for CRC screening programmes, and benchmarks should be determined. CRC screening uptake and adherence are the most crucial factors for the success of any screening programme. Studies on improving the health-seeking behaviour of asymptomatic individuals is another important subject of research in the Asia-Pacific region (online supplemental appendix 1).

Author affiliations
[1]Lee Kong Chian School of Medicine, Singapore
[2]Department of Internal Medicine, National Taiwan University Hospital, Taipei, Taiwan
[3]Department of Internal Medicine, College of Medicine, National Taiwan University, Taipei, Taiwan
[4]Oregon Health and Science University, Portland, Oregon, USA
[5]Erasmus MC, Rotterdam, The Netherlands
[6]University Hospital of North Tees, Stockton-on-Tees, UK
[7]The Royal Melbourne Hospital, Melbourne, Victoria, Australia
[8]National University of Singapore, Singapore
[9]Changi General Hospital, SingHealth Group, Singapore
[10]Raja Isteri Pengiran Anak Saleha Hospital, Brunei, Brunei Darussalam
[11]Digestive Health, Endoscopy, Gold Coast University Hospital, Southport, Queensland, Australia
[12]Peking Union Medical College Hospital, Beijing, China
[13]Fourth Military Medical University, Xi'an, China
[14]The Chinese University of Hong Kong, Hong Kong, Hong Kong
[15]All India Institute of Medical Sciences, New Delhi, Delhi, India
[16]Division of Gastroenterology, Pancreatibiliar and Digestive Endoscopy. Department of Internal Medicine, Hospital Dr Cipto Mangunkusumo, Jakarta, Indonesia
[17]Human Cancer Research Center. IMERI. Faculty of Medicine, Universitas Indonesia, Jakarta, Indonesia
[18]Cancer Screening Center/ Endoscopy Division, National Cancer Center Hospital, Tokyo, Japan
[19]Division of Screening Technology, National Cancer Center Institute for Cancer Control, Tokyo, Japan
[20]University of Ulsan College of Medicine, Seoul, Korea (the Republic of)
[21]Yonsei University, Seoul, Korea (the Republic of)
[22]National Bowel Screening Programme, New Zealand Ministry of Health, Auckland, New Zealand
[23]The University of Auckland, Auckland, New Zealand
[24]St Luke Hospital, Quezon City, Philippines
[25]Kaohsiung Medical University, Kaohsiung, Taiwan
[26]Chulalongkorn University, Bangkok, Thailand
[27]Division of Gastroenterology and Hepatology, Department of Medicine and Therapeutics, Prince of Wales Hospital, Chinese University of Hong Kong, Hong Kong, Hong Kong
[28]Institute of Digestive Disease, The Chinese University of Hong Kong, Hong Kong, Hong Kong
[29]Shin Kong Wu Ho Su Memorial Hospital, Taipei, Taiwan
[30]Amsterdam University Medical Centers, University of Amsterdam, Amsterdam, The Netherlands

Twitter Rashid N Lui @RashidLui and Sunny Wong @sunnyheiwong

Contributors 1. Coordinators (JJYS, H-MC, ED) were responsible for drafting the statements and coordinating the process of discussion and voting on the statements. 2. Advisors (DL, EJK, MDR, FM) helped to draft the first statements, moderate discussion during the meeting and revise the statements accordingly. 3. Panels (K-GY, TLA, VHC, JS, JL, KW, SSMN, GKM, MA, NK, MS, J-SB, H-SK, SP, PAIC-P, D-CW, SK) were country/region representatives who participated in the discussions and voting of the statements during the virtual conference. 4. Facilitators (SW, RNL, Y-ML) were responsible for performing the literature search, collating the discussion points and revising the statements.

Funding The authors have not received specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Not applicable.

Ethics approval Not applicable.

Provenance and peer review Not commissioned; externally peer reviewed.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

ORCID iDs
Joseph J Y Sung http://orcid.org/0000-0003-3125-5199
Han-Mo Chiu http://orcid.org/0000-0003-2786-8056
Ernst J Kuipers http://orcid.org/0000-0002-0633-3098
Matthew D Rutter http://orcid.org/0000-0001-9507-0295
Khay-Guan Yeoh http://orcid.org/0000-0002-7802-4606
Sneha John http://orcid.org/0000-0001-9332-3826
Govind K Makharia http://orcid.org/0000-0002-1305-189X
Nozomu Kobayashi http://orcid.org/0000-0003-2823-2375
Masau Sekiguchi http://orcid.org/0000-0001-6253-6896
Jeong-Sik Byeon http://orcid.org/0000-0002-9793-6379
Hyun-Soo Kim http://orcid.org/0000-0001-7190-0362
Rashid N Lui http://orcid.org/0000-0003-1277-9595
Sunny Wong http://orcid.org/0000-0002-3354-9310
E Dekker http://orcid.org/0000-0002-4363-0745

REFERENCES

1 Rabeneck L, Chiu H-M, Senore C. International perspective on the burden of colorectal cancer and public health effects. *Gastroenterology* 2020;158:447–52.
2 Arnold M, Sierra MS, Laversanne M, *et al*. Global patterns and trends in colorectal cancer incidence and mortality. *Gut* 2017;66:683–91.
3 Sung JJY, Lau JYW, Young GP, *et al*. Asia Pacific consensus recommendations for colorectal cancer screening. *Gut* 2008;57:1166–76.
4 Sung JJY, Ng SC, Chan FKL, *et al*. An updated Asia Pacific consensus recommendations on colorectal cancer screening. *Gut* 2015;64:121–32.
5 Onyoh EF, Hsu W-F, Chang L-C, *et al*. The rise of colorectal cancer in Asia: epidemiology, screening, and management. *Curr Gastroenterol Rep* 2019;21:36.
6 Pilonis ND, Bugajski M, Wieszczy P, *et al*. Participation in competing strategies for colorectal cancer screening: a randomized health services study (PICCOLINO study). *Gastroenterology* 2021;160:1097–105.
7 Chen H, Lu M, Liu C, *et al*. Comparative evaluation of participation and diagnostic yield of colonoscopy vs fecal immunochemical test vs risk-adapted screening in colorectal cancer screening: interim analysis of a multicenter randomized controlled trial (TARGET-C). *Am J Gastroenterol* 2020;115:1264–74.

8 Yeoh K-G, Ho K-Y, Chiu H-M, *et al*. The Asia-Pacific colorectal screening score: a validated tool that stratifies risk for colorectal advanced neoplasia in asymptomatic Asian subjects. *Gut* 2011;60:1236–41.
9 Sung JJY, Wong MCS, Lam TYT, *et al*. A modified colorectal screening score for prediction of advanced neoplasia: a prospective study of 5744 subjects. *J Gastroenterol Hepatol* 2018;33:187–94.
10 Corte C, Zhang L, Chen J, *et al*. Validation of the Asia Pacific colorectal screening (APCS) score in a Western population: an alternative screening tool. *J Gastroenterol Hepatol* 2016;31:370–5.
11 Aniwan S, Rerknimitr R, Kongkam P, *et al*. A combination of clinical risk stratification and fecal immunochemical test results to prioritize colonoscopy screening in asymptomatic participants. *Gastrointest Endosc* 2015;81:719–27.
12 Chiu H-M, Ching JYL, Wu KC, *et al*. A risk-scoring system combined with a fecal immunochemical test is effective in screening high-risk subjects for early colonoscopy to detect advanced colorectal neoplasms. *Gastroenterology* 2016;150:617–25.
13 Peng L, Balavarca Y, Weigl K, *et al*. Head-to-head comparison of the performance of 17 risk models for predicting presence of advanced neoplasms in colorectal cancer screening. *Am J Gastroenterol* 2019;114:1520–30.
14 Melson JE, Imperiale TF, Itzkowitz SH, *et al*. AGA white paper: roadmap for the future of colorectal cancer screening in the United States. *Clin Gastroenterol Hepatol* 2020;18:2667–78.
15 Zorzi M, Fedeli U, Schievano E, *et al*. Impact on colorectal cancer mortality of screening programmes based on the faecal immunochemical test. *Gut* 2015;64:784–90.
16 Chiu H-M, Jen GH-H, Wang Y-W, *et al*. Long-term effectiveness of faecal immunochemical test screening for proximal and distal colorectal cancers. *Gut* 2021;70:2321–9.
17 Wang K, Ma W, Wu K, *et al*. Long-term colorectal cancer incidence and mortality after colonoscopy screening according to individuals' risk profiles. *J Natl Cancer Inst* 2021;113:1177–85.
18 Jacob BJ, Moineddin R, Sutradhar R, *et al*. Effect of colonoscopy on colorectal cancer incidence and mortality: an instrumental variable analysis. *Gastrointest Endosc* 2012;76:355–64.
19 Nishihara R, Wu K, Lochhead P, *et al*. Long-term colorectal-cancer incidence and mortality after lower endoscopy. *N Engl J Med* 2013;369:1095–105.
20 Mandel JS, Bond JH, Church TR, *et al*. Reducing mortality from colorectal cancer by screening for fecal occult blood. Minnesota colon cancer control study. *N Engl J Med* 1993;328:1365–71.
21 Shaukat A, Mongin SJ, Geisser MS, *et al*. Long-term mortality after screening for colorectal cancer. *N Engl J Med* 2013;369:1106–14.
22 Knudsen AB, Rutter CM, Peterse EFP, *et al*. Colorectal cancer screening: an updated modeling study for the US Preventive Services Task force. *JAMA* 2021;325:1998–2011.
23 Lee JK, Jensen CD, Levin TR, *et al*. Long-term risk of colorectal cancer and related deaths after a colonoscopy with normal findings. *JAMA Intern Med* 2019;179:153–60.
24 Pilonis ND, Bugajski M, Wieszczy P, *et al*. Long-term colorectal cancer incidence and mortality after a single negative screening colonoscopy. *Ann Intern Med* 2020;173:81–91.
25 Atkin W, Wooldrage K, Parkin DM, *et al*. Long term effects of once-only flexible sigmoidoscopy screening after 17 years of follow-up: the UK flexible sigmoidoscopy screening randomised controlled trial. *Lancet* 2017;389:1299–311.
26 Holme Øyvind, Løberg M, Kalager M, *et al*. Long-Term effectiveness of sigmoidoscopy screening on colorectal cancer incidence and mortality in women and men: a randomized trial. *Ann Intern Med* 2018;168:775–82.
27 Senore C, Riggi E, Armaroli P. Long-Term follow-up of the Italian flexible sigmoidoscopy screening trial. *Ann Intern Med* 2022;175:36–45.
28 Sung JJ, Lau JY, Goh KL, Asia Pacific Working Group on Colorectal Cancer. Increasing incidence of colorectal cancer in Asia: implications for screening. *Lancet Oncol* 2005;6:871–6.
29 Imperiale TF, Ransohoff DF, Itzkowitz SH, *et al*. Multitarget stool DNA testing for colorectal-cancer screening. *N Engl J Med* 2014;370:1287–97.
30 Kwong GA, Ghosh S, Gamboa L, *et al*. Synthetic biomarkers: a twenty-first century path to early cancer detection. *Nat Rev Cancer* 2021;21:655–68.
31 Vuik FER, Nieuwenburg SAV, Moen S, *et al*. Colon capsule endoscopy in colorectal cancer screening: a systematic review. *Endoscopy* 2021;53:815–24.
32 Kim DH, Pickhardt PJ, Taylor AJ, *et al*. CT colonography versus colonoscopy for the detection of advanced neoplasia. *N Engl J Med* 2007;357:1403–12.
33 Stoop EM, de Haan MC, de Wijkerslooth TR, *et al*. Participation and yield of colonoscopy versus non-cathartic CT colonography in population-based screening for colorectal cancer: a randomised controlled trial. *Lancet Oncol* 2012;13:55–64.
34 Nagata K, Endo S, Honda T, *et al*. Accuracy of CT colonography for detection of polypoid and nonpolypoid neoplasia by gastroenterologists and radiologists: a nationwide multicenter study in Japan. *Am J Gastroenterol* 2017;112:163–71.
35 Hassan C, Pickhardt PJ. Cost-effectiveness of CT colonography. *Radiol Clin North Am* 2013;51:89–97.
36 Siegel RL, Miller KD, Jemal A. Colorectal cancer mortality rates in adults aged 20 to 54 years in the United States, 1970-2014. *JAMA* 2017;318:572–4.
37 Sung JJY, Chiu H-M, Jung K-W, *et al*. Increasing trend in young-onset colorectal cancer in Asia: more cancers in men and more rectal cancers. *Am J Gastroenterol* 2019;114:322–9.
38 Khil H, Kim SM, Hong S, *et al*. Time trends of colorectal cancer incidence and associated lifestyle factors in South Korea. *Sci Rep* 2021;11:2413.
39 Troeung L, Sodhi-Berry N, Martini A, *et al*. Increasing incidence of colorectal cancer in adolescents and young adults aged 15-39 years in Western Australia 1982-2007: examination of colonoscopy history. *Front Public Health* 2017;5:179.
40 Mathew A, Baby B, Wang K, *et al*. Colorectal cancer incidence in younger adults in India. *Gut* 2020;69:1899–900.
41 Chung RY-N, Tsoi KKF, Kyaw MH, *et al*. A population-based age-period-cohort study of colorectal cancer incidence comparing Asia against the West. *Cancer Epidemiol* 2019;59:29–36.
42 Siegel RL, Torre LA, Soerjomataram I, *et al*. Global patterns and trends in colorectal cancer incidence in young adults. *Gut* 2019;68:2179–85.
43 Gausman V, Dornblaser D, Anand S, *et al*. Risk factors associated with early-onset colorectal cancer. *Clin Gastroenterol Hepatol* 2020;18:2752–9.
44 O'Sullivan DE, Sutherland RL, Town S, *et al*. Risk factors for early-onset colorectal cancer: a systematic review and meta-analysis. *Clin Gastroenterol Hepatol* 2022;20:1229–40.
45 Joh H-K, Lee DH, Hur J, *et al*. Simple sugar and sugar-sweetened beverage intake during adolescence and risk of colorectal cancer precursors. *Gastroenterology* 2021;161:128–42.
46 Patel SG, Boland CR. Colorectal cancer in persons under age 50: seeking causes and solutions. *Gastrointest Endosc Clin N Am* 2020;30:441–55.
47 Rex DK, Boland CR, Dominitz JA, *et al*. Colorectal cancer screening: recommendations for physicians and patients from the U.S. Multi-Society Task Force on Colorectal Cancer. *Am J Gastroenterol* 2017;112:1016–30.
48 Wolf AMD, Fontham ETH, Church TR, *et al*. Colorectal cancer screening for average-risk adults: 2018 guideline update from the American Cancer Society. *CA Cancer J Clin* 2018;68:250–81.
49 US Preventive Services Task Force, Davidson KW, Barry MJ, *et al*. Screening for colorectal cancer: US Preventive Services Task Force recommendation statement. *JAMA* 2021;325:1965–77.
50 Patel SG, May FP, Anderson JC, *et al*. Updates on age to start and stop colorectal cancer screening: recommendations from the U.S. Multi-Society Task Force on Colorectal Cancer. *Gastroenterology* 2022;162:285–99.
51 Scholefield JH, Moss SM, Mangham CM, *et al*. Nottingham trial of faecal occult blood testing for colorectal cancer: a 20-year follow-up. *Gut* 2012;61:1036–40.
52 Faivre J, Dancourt V, Lejeune C, *et al*. Reduction in colorectal cancer mortality by fecal occult blood screening in a French controlled study. *Gastroenterology* 2004;126:1674–80.
53 Ladabaum U, Mannalithara A, Meester RGS, *et al*. Cost-effectiveness and national effects of initiating colorectal cancer screening for average-risk persons at age 45 years instead of 50 years. *Gastroenterology* 2019;157:137–48.
54 Lew J-B, St John DJB, Macrae FA, *et al*. Benefits, harms, and cost-effectiveness of potential age extensions to the National Bowel Cancer Screening Program in Australia. *Cancer Epidemiol Biomarkers Prev* 2018;27:1450–61.
55 van Hees F, Saini SD, Lansdorp-Vogelaar I, *et al*. Personalizing colonoscopy screening for elderly individuals based on screening history, cancer risk, and comorbidity status could increase cost effectiveness. *Gastroenterology* 2015;149:1425–37.
56 Kastrinos F, Samadder NJ, Burt RW. Use of family history and genetic testing to determine risk of colorectal cancer. *Gastroenterology* 2020;158:389–403.
57 Uson PLS, Riegert-Johnson D, Boardman L, *et al*. Germline cancer susceptibility gene testing in unselected patients with colorectal adenocarcinoma: a multicenter prospective study. *Clin Gastroenterol Hepatol* 2022;20:e508–28.
58 Gupta S, Bharti B, Ahnen DJ, *et al*. Potential impact of family history-based screening guidelines on the detection of early-onset colorectal cancer. *Cancer* 2020;126:3013–20.
59 Leddin D, Lieberman DA, Tse F, *et al*. Clinical practice guideline on screening for colorectal cancer in individuals with a family history of nonhereditary colorectal cancer or adenoma: the Canadian Association of Gastroenterology Banff consensus. *Gastroenterology* 2018;155:1325–47.
60 Ahadova A, Seppälä TT, Engel C, *et al*. The "unnatural" history of colorectal cancer in Lynch syndrome: lessons from colonoscopy surveillance. *Int J Cancer* 2021;148:800–11.
61 Sung JJY, Choi SYP, Chan FKL, *et al*. Obstacles to colorectal cancer screening in Chinese: a study based on the health belief model. *Am J Gastroenterol* 2008;103:974–81.
62 Ng SC, Zeng Z, Niewiadomski O, *et al*. Early course of inflammatory bowel disease in a population-based inception cohort study from 8 countries in Asia and Australia. *Gastroenterology* 2016;150:86–95.
63 Ng SC, Kyaw MH, Suen BY, *et al*. Prospective colonoscopic study to investigate risk of colorectal neoplasms in first-degree relatives of patients with non-advanced adenomas. *Gut* 2020;69:304–10.

64 Samadder NJ, Curtin K, Tuohy TMF, *et al*. Increased risk of colorectal neoplasia among family members of patients with colorectal cancer: a population-based study in Utah. *Gastroenterology* 2014;147:814–21.
65 Chung SJ, Kim YS, Yang SY, *et al*. Five-year risk for advanced colorectal neoplasia after initial colonoscopy according to the baseline risk stratification: a prospective study in 2452 asymptomatic Koreans. *Gut* 2011;60:1537–43.
66 Chiu H-M, Lee Y-C, Tu C-H, *et al*. Effects of metabolic syndrome and findings from baseline colonoscopies on occurrence of colorectal neoplasms. *Clin Gastroenterol Hepatol* 2015;13:1134–42.
67 Hassan C, Antonelli G, Dumonceau J-M, *et al*. Post-polypectomy colonoscopy surveillance: European Society of Gastrointestinal Endoscopy (ESGE) Guideline - Update 2020. *Endoscopy* 2020;52:687–700.
68 Gupta S. Cumulative risk for incident and fatal colorectal cancer after polypectomy. *Gastroenterology* 2020;159:1992.
69 Lieberman DA, Williams JL, Holub JL, *et al*. Colonoscopy utilization and outcomes 2000 to 2011. *Gastrointest Endosc* 2014;80:133–43.
70 Kaminski MF, Regula J, Kraszewska E, *et al*. Quality indicators for colonoscopy and the risk of interval cancer. *N Engl J Med* 2010;362:1795–803.
71 Corley DA, Jensen CD, Marks AR, *et al*. Adenoma detection rate and risk of colorectal cancer and death. *N Engl J Med* 2014;370:1298–306.
72 Chiu SY-H, Chuang S-L, Chen SL-S, *et al*. Faecal haemoglobin concentration influences risk prediction of interval cancers resulting from inadequate colonoscopy quality: analysis of the Taiwanese nationwide colorectal cancer screening program. *Gut* 2017;66:293–300.
73 Pabby A, Schoen RE, Weissfeld JL, *et al*. Analysis of colorectal cancer occurrence during surveillance colonoscopy in the dietary polyp prevention trial. *Gastrointest Endosc* 2005;61:385–91.
74 le Clercq CMC, Bouwens MWE, Rondagh EJA, *et al*. Postcolonoscopy colorectal cancers are preventable: a population-based study. *Gut* 2014;63:957–63.
75 Click B, Pinsky PF, Hickey T, *et al*. Association of colonoscopy adenoma findings with long-term colorectal cancer incidence. *JAMA* 2018;319:2021–31.
76 He X, Hang D, Wu K, *et al*. Long-term risk of colorectal cancer after removal of conventional adenomas and serrated polyps. *Gastroenterology* 2020;158:852–61.
77 Wieszczy P, Kaminski MF, Franczyk R, *et al*. Colorectal cancer incidence and mortality after removal of adenomas during screening colonoscopies. *Gastroenterology* 2020;158:875–83.
78 Lee JK, Jensen CD, Levin TR, *et al*. Long-term risk of colorectal cancer and related death after adenoma removal in a large, community-based population. *Gastroenterology* 2020;158:884–94.
79 Jover R, Dekker E, Schoen RE, *et al*. Colonoscopy quality requisites for selecting surveillance intervals: a World Endoscopy Organization Delphi recommendation. *Dig Endosc* 2018;30:750–9.
80 Robertson DJ, Lieberman DA, Winawer SJ, *et al*. Colorectal cancers soon after colonoscopy: a pooled multicohort analysis. *Gut* 2014;63:949–56.
81 Pohl H, Anderson JC, Aguilera-Fish A, *et al*. Recurrence of colorectal neoplastic polyps after incomplete resection. *Ann Intern Med* 2021;174:1377–84.
82 Pinsky PF, Schoen RE. Contribution of surveillance colonoscopy to colorectal cancer prevention. *Clin Gastroenterol Hepatol* 2020;18:2937–44.
83 Schoen RE, Pinsky PF, Weissfeld JL, *et al*. Utilization of surveillance colonoscopy in community practice. *Gastroenterology* 2010;138:73–81.
84 Oh CK, Aniwan S, Piyachaturawat P, *et al*. Adherence to surveillance guidelines after the removal of colorectal polyps: a multinational, multicenter, prospective survey. *Gut Liver* 2021;15:878–86.
85 van Heijningen E-MB, Lansdorp-Vogelaar I, Steyerberg EW, *et al*. Adherence to surveillance guidelines after removal of colorectal adenomas: a large, community-based study. *Gut* 2015;64:1584–92.
86 Matsuda T, Chiu H-M, Sano Y, *et al*. Surveillance colonoscopy after endoscopic treatment for colorectal neoplasia: from the standpoint of the Asia-Pacific region. *Dig Endosc* 2016;28:342–7.
87 Duvvuri A, Chandrasekar VT, Srinivasan S, *et al*. Risk of colorectal cancer and cancer related mortality after detection of low-risk or high-risk adenomas, compared with no adenoma, at index colonoscopy: a systematic review and meta-analysis. *Gastroenterology* 2021;160:1986–96.
88 Dubé C, Yakubu M, McCurdy BR, *et al*. Risk of advanced adenoma, colorectal cancer, and colorectal cancer mortality in people with low-risk adenomas at baseline colonoscopy: a systematic review and meta-analysis. *Am J Gastroenterol* 2017;112:1790–801.
89 Kuntz KM, Lansdorp-Vogelaar I, Rutter CM, *et al*. A systematic comparison of microsimulation models of colorectal cancer: the role of assumptions about adenoma progression. *Med Decis Making* 2011;31:530–9.
90 Brenner H, Altenhofen L, Stock C, *et al*. Natural history of colorectal adenomas: birth cohort analysis among 3.6 million participants of screening colonoscopy. *Cancer Epidemiol Biomarkers Prev* 2013;22:1043–51.
91 Winawer SJ, Zauber AG, O'Brien MJ, *et al*. Randomized comparison of surveillance intervals after colonoscopic removal of newly diagnosed adenomatous polyps. The National Polyp Study Workgroup. *N Engl J Med* 1993;328:901–6.
92 Matsuda T, Fujii T, Sano Y. Randomised comparison of postpolypectomy surveillance intervals following a two-round baseline colonoscopy: the Japan Polyp Study Workgroup. *Gut* 2021;70:1469.
93 Saito Y, Oka S, Kawamura T, *et al*. Colonoscopy screening and surveillance guidelines. *Dig Endosc* 2021;33:486–519.
94 Yang D-H, Hong SN, Kim Y-H, *et al*. Korean guidelines for postpolypectomy colonoscopy surveillance. *Clin Endosc* 2012;45:44–61.
95 Peng S-M, Hsu W-F, Wang Y-W, *et al*. Faecal immunochemical test after negative colonoscopy may reduce the risk of incident colorectal cancer in a population-based screening programme. *Gut* 2021;70:1318–24.
96 Cross AJ, Wooldrage K, Robbins EC, *et al*. Faecal immunochemical tests (FIT) versus colonoscopy for surveillance after screening and polypectomy: a diagnostic accuracy and cost-effectiveness study. *Gut* 2019;68:1642–52.
97 Bogie RMM, le Clercq CMC, Voorham QJM, *et al*. Molecular pathways in post-colonoscopy versus detected colorectal cancers: results from a nested case-control study. *Br J Cancer* 2022;126:865–73.
98 Bosman FT CF, Hruban RH, Theise ND. *Who classification of tumours of the digestive system*. 4th Edition, Volume 3. International Agency for Research on Cancer, 2010.
99 WHO Classification of Tumours Editorial Board. *Who classification of tumours: digestive system tumours*. 5th Edition, Volume 1. International Agency for Research on Cancer, 2019.
100 van Toledo DEFWM, IJspeert JEG, Bossuyt PMM, *et al*. Serrated polyp detection and risk of interval post-colonoscopy colorectal cancer: a population-based study. *Lancet Gastroenterol Hepatol* 2022;7:747-754.
101 East JE, Atkin WS, Bateman AC, *et al*. British Society of Gastroenterology position statement on serrated polyps in the colon and rectum. *Gut* 2017;66:1181–96.
102 Samadder NJ, Neklason D, Snow A, *et al*. Clinical and molecular features of post-colonoscopy colorectal cancers. *Clin Gastroenterol Hepatol* 2019;17:2731–9.
103 Bettington M, Walker N, Rahman T, *et al*. High prevalence of sessile serrated adenomas in contemporary outpatient colonoscopy practice. *Intern Med J* 2017;47:318–23.
104 Crockett SD, Terminology NID, Features M. Epidemiology, and management of serrated colorectal neoplasia. *Gastroenterology* 2019;157:949–66.
105 Song M, Emilsson L, Bozorg SR, *et al*. Risk of colorectal cancer incidence and mortality after polypectomy: a Swedish record-linkage study. *Lancet Gastroenterol Hepatol* 2020;5:537–47.
106 Meester RGS, van Herk MMAGC, Lansdorp-Vogelaar I, *et al*. Prevalence and clinical features of sessile serrated polyps: a systematic review. *Gastroenterology* 2020;159:105–18.
107 Sekiguchi M, Kakugawa Y, Matsumoto M, *et al*. Prevalence of serrated lesions, risk factors, and their association with synchronous advanced colorectal neoplasia in asymptomatic screened individuals. *J Gastroenterol Hepatol* 2020;35:1938–44.
108 Lui RN, Kyaw MH, Lam TYT, *et al*. Prevalence and risk factors for sessile serrated lesions in an average risk colorectal cancer screening population. *J Gastroenterol Hepatol* 2021;36:1656–62.
109 Liu TY, Jin DC, Khan S, *et al*. Clinicopathological features of advanced colorectal serrated lesions: a single-center study in China. *J Dig Dis* 2018;19:235–41.
110 Kumbhari V, Behary J, Hui JM. Prevalence of adenomas and sessile serrated adenomas in Chinese compared with Caucasians. *J Gastroenterol Hepatol* 2013;28:608–12.
111 Kahi CJ, Hewett DG, Norton DL, *et al*. Prevalence and variable detection of proximal colon serrated polyps during screening colonoscopy. *Clin Gastroenterol Hepatol* 2011;9:42–6.
112 Desai M, Anderson JC, Kaminski M, *et al*. Sessile serrated lesion detection rates during average risk screening colonoscopy: a systematic review and meta-analysis of the published literature. *Endosc Int Open* 2021;9:E610–20.
113 Huang J, Chan PSF, Pang TWY, *et al*. Rate of detection of serrated lesions at colonoscopy in an average-risk population: a meta-analysis of 129,001 individuals. *Endosc Int Open* 2021;9:E472–81.
114 Gill P, Rafferty H, Munday D, *et al*. Proximal colon cancer and serrated adenomas - hunting the missing 10%. *Clin Med* 2013;13:557–61.
115 Lin Y-C, Chiu H-M, Lee Y-C, *et al*. Hyperplastic polyps identified during screening endoscopy: reevaluated by histological examinations and genetic alterations. *J Formos Med Assoc* 2014;113:417–21.
116 Tadepalli US, Feihel D, Miller KM, *et al*. A morphologic analysis of sessile serrated polyps observed during routine colonoscopy (with video). *Gastrointest Endosc* 2011;74:1360–8.
117 Hazewinkel Y, López-Cerón M, East JE, *et al*. Endoscopic features of sessile serrated adenomas: validation by international experts using high-resolution white-light endoscopy and narrow-band imaging. *Gastrointest Endosc* 2013;77:916–24.
118 IJspeert JEG, Bastiaansen BAJ, van Leerdam ME, *et al*. Development and validation of the WASP classification system for optical diagnosis of adenomas, hyperplastic polyps and sessile serrated adenomas/polyps. *Gut* 2016;65:963–70.
119 Bleijenberg AGC, van Leerdam ME, Bargeman M, *et al*. Substantial and sustained improvement of serrated polyp detection after a simple educational intervention: results from a prospective controlled trial. *Gut* 2020;69:2150–8.

120 Rutter MD, Beintaris I, Valori R, *et al*. World Endoscopy Organization consensus statements on post-colonoscopy and post-imaging colorectal cancer. *Gastroenterology* 2018;155:909–25.
121 Morris EJA, Rutter MD, Finan PJ, *et al*. Post-colonoscopy colorectal cancer (PCCRC) rates vary considerably depending on the method used to calculate them: a retrospective observational population-based study of PCCRC in the English National Health Service. *Gut* 2015;64:1248–56.
122 Baxter NN, Goldwasser MA, Paszat LF, *et al*. Association of colonoscopy and death from colorectal cancer. *Ann Intern Med* 2009;150:1–8.
123 Arain MA, Sawhney M, Sheikh S, *et al*. CIMP status of interval colon cancers: another piece to the puzzle. *Am J Gastroenterol* 2010;105:1189–95.
124 Erichsen R, Baron JA, Hamilton-Dutoit SJ, *et al*. Increased risk of colorectal cancer development among patients with serrated polyps. *Gastroenterology* 2016;150:895–902.
125 Anderson JC, Butterly LF, Robinson CM, *et al*. Risk of metachronous high-risk adenomas and large serrated polyps in individuals with serrated polyps on index colonoscopy: data from the new Hampshire Colonoscopy Registry. *Gastroenterology* 2018;154:117–27.
126 Sano W, Fujimori T, Ichikawa K, *et al*. Clinical and endoscopic evaluations of sessile serrated adenoma/polyps with cytological dysplasia. *J Gastroenterol Hepatol* 2018;33:1454–60.
127 Holme Øyvind, Bretthauer M, Eide TJ, *et al*. Long-term risk of colorectal cancer in individuals with serrated polyps. *Gut* 2015;64:929–36.
128 van Hattem WA, Shahidi N, Vosko S, *et al*. Piecemeal cold SNARE polypectomy versus conventional endoscopic mucosal resection for large sessile serrated lesions: a retrospective comparison across two successive periods. *Gut* 2021;70:1691–7.
129 Yoshida N, Inoue K, Tomita Y, *et al*. Cold SNARE polypectomy for large sessile serrated lesions is safe but follow-up is needed: a single-centre retrospective study. *United European Gastroenterol J* 2021;9:370–7.
130 Rao AK, Soetikno R, Raju GS, *et al*. Large sessile serrated polyps can be safely and effectively removed by endoscopic mucosal resection. *Clin Gastroenterol Hepatol* 2016;14:568–74.
131 Gao Q, Tsoi KKF, Hirai HW, *et al*. Serrated polyps and the risk of synchronous colorectal advanced neoplasia: a systematic review and meta-analysis. *Am J Gastroenterol* 2015;110:501–9.
132 Segnan NPJ, von Karsa L. European guidelines for quality assurance in colorectal cancer screening. *1st Edition* 2010.
133 Zorzi M, Battagello J, Selby K, *et al*. Non-compliance with colonoscopy after a positive faecal immunochemical test doubles the risk of dying from colorectal cancer. *Gut* 2022;71:561–7.
134 Australian Institute of Health and Welfare. National bowel cancer screening program: monitoring report 2021, 2021. Available: https://www.aihw.gov.au/reports/cancer-screening/nbcsp-monitoring-report-2021/summary [Accessed 28 Jan 2022].
135 Japan Ministry of Health Labour and Welfare. Health of household members. *Comprehensive Survey of Living Conditions* 2019;2019 https://www.mhlw.go.jp/toukei/saikin/hw/k-tyosa/k-tyosa19/dl/04.pdf
136 Hong S, Lee YY, Lee J, *et al*. Trends in cancer screening rates among Korean men and women: results of the Korean National Cancer Screening Survey, 2004-2018. *Cancer Res Treat* 2021;53:330–8.
137 Choi KS, Lee H-Y, Jun JK, *et al*. Adherence to follow-up after a positive fecal occult blood test in an organized colorectal cancer screening program in Korea, 2004-2008. *J Gastroenterol Hepatol* 2012;27:1070–7.
138 Lee Y-C, Li-Sheng Chen S, Ming-Fang Yen A, *et al*. Association between colorectal cancer mortality and gradient fecal hemoglobin concentration in colonoscopy noncompliers. *J Natl Cancer Inst* 2017;109. doi:10.1093/jnci/djw269. [Epub ahead of print: 01 05 2017].
139 Netherlands Comprehensive Cancer Organisation. National monitoring colorectal cancer screening programme 2020, 2021. Available: https://www.rivm.nl/en/media/199021 [Accessed 28 Jan 2022].
140 Hsu W-F, Cheng S-Y, Shun C-T, *et al*. Higher risk of advanced histology in adenoma less than 10 mm in fecal immunochemical test screening: Implication for management. *J Gastroenterol Hepatol* 2020;35:1738–45.
141 Katsoula A, Paschos P, Haidich A-B, *et al*. Diagnostic accuracy of fecal immunochemical test in patients at increased risk for colorectal cancer: a meta-analysis. *JAMA Intern Med* 2017;177:1110–8.
142 Lu M, Wang L, Zhang Y, *et al*. Optimizing positivity thresholds for a risk-adapted screening strategy in colorectal cancer screening. *Clin Transl Gastroenterol* 2021;12:e00398.
143 San Miguel Y, Demb J, Martinez ME, *et al*. Time to colonoscopy after abnormal stool-based screening and risk for colorectal cancer incidence and mortality. *Gastroenterology* 2021;160:1997–2005.
144 Lee Y-C, Fann JC-Y, Chiang T-H, *et al*. Time to colonoscopy and risk of colorectal cancer in patients with positive results from fecal immunochemical tests. *Clin Gastroenterol Hepatol* 2019;17:1332–40.
145 Corley DA, Jensen CD, Quinn VP, *et al*. Association between time to colonoscopy after a positive fecal test result and risk of colorectal cancer and cancer stage at diagnosis. *JAMA* 2017;317:1631–41.
146 Zorzi M, Hassan C, Capodaglio G, *et al*. Colonoscopy later than 270 days in a fecal immunochemical test-based population screening program is associated with higher prevalence of colorectal cancer. *Endoscopy* 2020;52:871–6.
147 de Jonge L, Worthington J, van Wifferen F, *et al*. Impact of the COVID-19 pandemic on faecal immunochemical test-based colorectal cancer screening programmes in Australia, Canada, and the Netherlands: a comparative modelling study. *Lancet Gastroenterol Hepatol* 2021;6:304–14.
148 Dekker E, Chiu H-M, Lansdorp-Vogelaar I, *et al*. Colorectal cancer screening in the novel coronavirus disease-2019 era. *Gastroenterology* 2020;159:1998–2003.
149 Chiu H-M, Su C-W, Hsu W-F, *et al*. Mitigating the impact of COVID-19 on colorectal cancer screening: organized service screening perspectives from the Asia-Pacific region. *Prev Med* 2021;151:106622.
150 Honein-AbouHaidar GN, Kastner M, Vuong V, *et al*. Systematic review and meta-study synthesis of qualitative studies evaluating facilitators and barriers to participation in colorectal cancer screening. *Cancer Epidemiol Biomarkers Prev* 2016;25:907–17.
151 Koo JH, Leong RWL, Ching J, *et al*. Knowledge of, attitudes toward, and barriers to participation of colorectal cancer screening tests in the Asia-Pacific region: a multicenter study. *Gastrointest Endosc* 2012;76:126–35.
152 Yusoff HM, Daud N, Noor NM, *et al*. Participation and barriers to colorectal cancer screening in Malaysia. *Asian Pac J Cancer Prev* 2012;13:3983–7.
153 Azulay R, Valinsky L, Hershkowitz F, *et al*. Barriers to completing colonoscopy after a positive fecal occult blood test. *Isr J Health Policy Res* 2021;10:11.
154 Cheng S-Y, Li M-C, Chia S-L, *et al*. Factors affecting compliance with confirmatory colonoscopy after a positive fecal immunochemical test in a national colorectal screening program. *Cancer* 2018;124:907–15.
155 Gies A, Cuk K, Schrotz-King P, *et al*. Direct comparison of diagnostic performance of 9 quantitative fecal immunochemical tests for colorectal cancer screening. *Gastroenterology* 2018;154:93–104.
156 Chiang T-H, Chuang S-L, Chen SL-S, *et al*. Difference in performance of fecal immunochemical tests with the same hemoglobin cutoff concentration in a nationwide colorectal cancer screening program. *Gastroenterology* 2014;147:1317–26.
157 Cha JM, Suh M, Kwak MS, *et al*. Risk of interval cancer in fecal immunochemical test screening significantly higher during the summer months: results from the National Cancer Screening Program in Korea. *Am J Gastroenterol* 2018;113:611–21.
158 Burr NE, Derbyshire E, Taylor J, *et al*. Variation in post-colonoscopy colorectal cancer across colonoscopy providers in English National Health Service: population based cohort study. *BMJ* 2019;367:l6090.
159 Kaminski MF, Wieszczy P, Rupinski M, *et al*. Increased rate of adenoma detection associates with reduced risk of colorectal cancer and death. *Gastroenterology* 2017;153:98–105.
160 Wong JCT, Chiu H-M, Kim H-S, *et al*. Adenoma detection rates in colonoscopies for positive fecal immunochemical tests versus direct screening colonoscopies. *Gastrointest Endosc* 2019;89:607–13.
161 Bronzwaer MES, Depla ACTM, van Lelyveld N, *et al*. Quality assurance of colonoscopy within the Dutch national colorectal cancer screening program. *Gastrointest Endosc* 2019;89:1–13.
162 Provenzale D, Ness RM, Llor X, *et al*. NCCN guidelines insights: colorectal cancer screening, version 2.2020. *J Natl Compr Canc Netw* 2020;18:1312–20.
163 Helwick C. Changes in the NCCN guidelines on colorectal cancer screening, 2022. Available: https://ascopost.com/news/april-2022/changes-in-the-nccn-guidelines-on-colorectal-cancer-screening/ [Accessed 11 May 2022].
164 Shaukat A, Kahi CJ, Burke CA, *et al*. ACG clinical guidelines: colorectal cancer screening 2021. *Am J Gastroenterol* 2021;116:458–79.
165 Jenkins MA, Ait Ouakrim D, Boussioutas A, *et al*. Revised Australian national guidelines for colorectal cancer screening: family history. *Med J Aust* 2018;209:455–60.
166 van Leerdam ME, Roos VH, van Hooft JE, *et al*. Endoscopic management of Lynch syndrome and of familial risk of colorectal cancer: European Society of Gastrointestinal Endoscopy (ESGE) guideline. *Endoscopy* 2019;51:1082–93.
167 Monahan KJ, Bradshaw N, Dolwani S, *et al*. Guidelines for the management of hereditary colorectal cancer from the British Society of Gastroenterology (BSG)/Association of Coloproctology of Great Britain and Ireland (ACPGBI)/United Kingdom Cancer Genetics Group (UKCGG). *Gut* 2020;69:411–44.